



EVALUATING THE UTILITY OF SYNTHETIC T₁C BRAIN MRI SCANS GENERATED BY GAN AND DIFFUSION MODELS

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Abstract

Glioma is the most common malignant brain tumor, carrying a poor prognosis. T₁-weighted post-contrast (T₁c) MRI scans are essential for accurate tumor grading and subregion delineation. Limited scanner availability and contraindications make these scans frequently absent, negatively impacting patient care and segmentation models, which rely on complete data. Generative models have been applied to missing MRI synthesis, but T₁c remains underexplored.

This thesis addresses two questions: to what extent can state-of-the-art generative models synthesize high-fidelity 2D T₁c images, and to what extent can synthetic T₁c substitute for missing T₁c in glioma segmentation. One unconditional (StyleGAN2-ADA) and three conditional models (pix2pixRAD, SynDiff, and a custom diffusion model) were trained and assessed on the BraTS-2023 Adult Glioma dataset with generalization tested on the UCSF-PDGM dataset.

Image fidelity was assessed along three dimensions, including distributional realism (FID) and structural similarity to ground truth (SSIM) as representative metrics. High-fidelity T₁c were synthesized by SynDiff (FID: 18.52; SSIM: 0.9415) and pix2pixRAD (FID: 37.17; SSIM: 0.9509), while custom diffusion failed (FID: 97.51; SSIM: 0.1061). SynDiff generalized better than pix2pixRAD (e.g., FID: 19.35 vs. 48.83). StyleGAN2-ADA learned the T₁c distribution (FID: 20.44) but struggled to generalize (FID: 68.75).

Segmentation accuracy, measured via Dice score (overlap with real tumor masks), improved with pix2pixRAD and SynDiff substitution (e.g., 50%: 0.7011 & 0.6770) over missing T₁c conditions (e.g., 50% missing: 0.4794), with similar UCSF-PDGM gains. Enhancing tumor regions and

small tumors underperformed, indicating disparate clinical impact. Partial synthetic T1c clinical utility is supported, with enhancing tumor learning a remaining challenge.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

The primary data used in this publication were obtained as part of the Brain Tumor Segmentation (BraTS) Challenge project through Synapse ID: syn51156910 (Baid et al., 2021; Bakas et al., 2017a, 2017b, 2017c; Menze et al., 2015). Data were collected and compiled with consent from adult glioma patients at various hospitals. This study also used the images and annotations subset of the UCSF-PDGM dataset (Calabrese et al., 2023), which is hosted by the Cancer Imaging Archive (Clark et al., 2013) and licensed under CC by 4.0. The use of BraTS 2023 Adult Glioma data and the UCSF-PDGM data in this project is consistent with the Attribution-NonCommercial 4.0 International deed, as it is used strictly for research and not intended for commercial use. All data were previously anonymized. Both datasets primarily represent western demographics. No further image data was collected or acquired. Aside from the use of BraTS and UCSF-PDGM data for comparison in several figures, as well as images generated using generative models cited below, all images belong to the author.

All model codes were sourced via open-source repositories provided by the authors (Conte et al., 2021; Isensee et al., 2021; Karras et al., 2020; Mirowski & Fabijańska, 2026; Özbey et al., 2023). Code was adapted as necessary to suit the needs of the experiments conducted in this study. The repositories and corresponding original code are provided in the footnotes.¹
² ³ ⁴ ⁵. Data preprocessing was conducted according to the individual needs of the generative models. Preprocessing and analysis code used for this thesis is made publicly available ⁶.

The main Python libraries used in this study were PyTorch/torchvision (Paszke et al., 2019), MONAI (Cardoso et al., 2022), Scikit-learn (Pedregosa et al., 2011), Scikit-image (Walt et al., 2014), Matplotlib (Hunter, 2007), TensorFlow (Abadi et al., 2016), NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), and Pandas (McKinney, 2010). Full libraries and associated versions are available in [Appendix A](#), Table 11.

Microsoft Word spellcheck was used to correct spelling errors. The thesis was adapted to and presented in Overleaf. References were managed

¹ <https://github.com/giemmecci/pix2pixRAD>

² <https://github.com/NVlabs/stylegan2-ada-pytorch>

³ <https://github.com/icon-lab/SynDiff>

⁴ <https://github.com/Miretpl/synthetic-brain-mri-project>

⁵ <https://github.com/MIC-DKFZ/nnUNet>

⁶ <https://github.com/jolbinsk1/Brain-MRI-Generation-and-Segmentation>

with Zotero. Draw.io was used in creating figures. Claude.ai (Anthropic, 2024) was used to aid in the construction of code, presentation of figures and tables, as well as improve writing. AI-generated text was revised and integrated by the author.

2 PROBLEM STATEMENT AND RELEVANCE

2.1 Problem Statement

This study addresses the problem that T1c scans, critical for glioma assessment, are often missing, adversely affecting patient care and clinical research. The central goal is to investigate to what extent state-of-the-art generative models can be adapted to synthesize realistic T1c capable of substituting for missing scans in downstream glioma segmentation.

2.2 Clinical Burden of Brain Tumors

Brain tumors are often highly aggressive, with an average 5-year survival rate of roughly 35% for malignant tumors (Ostrom et al., 2022). Of these, gliomas are the most common, representing roughly 81% of cases (Ostrom et al., 2014, 2022). Gliomas are classified by the WHO according to severity, with low-grade (LGG) corresponding to grades I–II, and high-grade (HGG) to grades III–IV (Louis et al., 2021). Glioblastomas, a subtype of HGG, are the most malignant and common glioma subtype, representing 40–50% of total cases and carrying a median overall survival of 14–16 months and a 5-year survival rate of below 10% (Aleid et al., 2024; Ostrom et al., 2022).

Nonetheless, outcomes are improved with early diagnosis and treatment (Thenuwara et al., 2023). Magnetic resonance imaging (MRI) has emerged as an indispensable tool for non-invasive brain tumor diagnosis, assessment, and segmentation (Falk Delgado, 2025; Wei & Wei, 2021); and MRI-trained segmentation models have helped improve tumor segmentation accuracy and reduce time and effort (Isensee et al., 2021; Laukamp et al., 2021).

2.3 T1c Scan Importance and Missingness

Brain MRI scans are multi-modal, encompassing T1-weighted (T1w), T2-weighted (T2w), Fluid-Attenuated Inversion Recovery (FLAIR), and T1-weighted post-contrast (T1c) scans, and are often examined together for accurate clinical assessment (Baid et al., 2021; Kim & Park, 2024). Among these, T1c scans are widely used in glioma imaging to properly delineate

glioma grade and anatomy (e.g., LGG/HGG differentiation) by using gadolinium to highlight sites of blood-brain barrier disruption (Hirschler et al., 2023; Jayachandran Preetha et al., 2021; Sarkaria et al., 2018; Warntjes et al., 2018).

Despite their importance, T1c scans are frequently absent from both clinical and research datasets due to limited scanner availability, differing protocols, scan time, and cost (Kim & Park, 2024; Nafi et al., 2024; Zhou et al., 2022). Conventional data augmentation methods cannot adequately compensate for sparse unique data, and missing T1c cannot be directly replaced by non-contrast modalities at inference time, motivating the use of generative models to synthesize missing modalities (Islam et al., 2026; L. Li et al., 2024; Nafi et al., 2024; Raut et al., 2024).

2.4 *Generative Models in Brain MRI*

Brain MRI synthesis is dominated by generative adversarial networks (GANs) and denoising diffusion probabilistic models (DDPMs, or diffusion models), which address scarcity by learning underlying image distributions to synthesize realistic images (Kazerouni et al., 2023; Nafi et al., 2024; Tavse et al., 2022). GANs consist of two competing networks: a generator, which produces synthetic images; and a discriminator, which attempts to distinguish real from synthetic images. These are trained jointly until the generator fools the discriminator (Goodfellow et al., 2014). However, they may be sensitive to training instability and may struggle to synthesize diverse images (Saad et al., 2024; Skandarani et al., 2023). Diffusion models instead learn to generate images by reversing a gradual noising process (Ho et al., 2020). While they have been found to produce more diverse, higher-fidelity brain MRI images than GANs (Dorjsembe et al., 2024; Müller-Franzes et al., 2023), they do so at a greater computational cost (Kazerouni et al., 2023; Usman Akbar et al., 2024).

Both can be categorized into two general types. Conditional models map existing scans onto a different target — e.g., generating T1w from T1c (Conte et al., 2021) — making them suitable for missing scan substitution (Raut et al., 2024). Unconditional models learn the distribution of a single target and generate novel images through random sampling, making them suited for dataset augmentation (Motamed et al., 2021; Tariq et al., 2023). Both types have been applied to 2D and 3D brain MRI synthesis (Conte et al., 2021; Friedrich et al., 2025; Özbey et al., 2023; Raut et al., 2024). Synthetic MRI quality is typically assessed via image fidelity metrics, though evaluation approaches vary considerably (Deo et al., 2025; Dohmen et al., 2025). Additionally, synthetic scans are increasingly assessed beyond

image fidelity for downstream clinical utility, particularly segmentation performance (Raut et al., 2024; Usman Akbar et al., 2024; Wu et al., 2024).

This study evaluates four state-of-the-art 2D representatives of these model families: a conditional GAN (*pix2pixRAD*; Conte et al. 2021), an unconditional GAN (*StyleGAN2-ADA*; Karras et al. 2020), and two conditional diffusion models (*SynDiff*; Özbey et al. 2023, and a custom diffusion model; Mirowski and Fabijańska 2026). Each has demonstrated high-fidelity non-contrast brain MRI synthesis: *pix2pixRAD* synthesized T1w and FLAIR for missing scan substitution (Conte et al., 2021; Zoghby et al., 2024); *StyleGAN2-ADA* synthesized healthy and glioma T1w (Lai et al., 2025; Tariq et al., 2023); *SynDiff* demonstrated bidirectional multi-modal synthesis (Özbey et al., 2023); while the custom diffusion model synthesized FLAIR from tumor segmentation masks (Mirowski & Fabijańska, 2026). However, none have been adapted to T1c synthesis.

2.5 Societal Motivation

The societal motivation behind this study is to contribute to positive glioma treatment outcomes. Absent T1c reduces the ability to characterize enhancement patterns common in HGGs and critical in treatment planning (Hirschler et al., 2023; Unkelbach et al., 2014). Additionally, gadolinium has several contraindications, including severe complications in kidney disease, fetal development risks in pregnancy, and potential long-term accumulation risks (Bird et al., 2019; Jayachandran Preetha et al., 2021; Schieda et al., 2018). Therefore, high-fidelity substitute T1c may help vulnerable patients overcome resource barriers and avoid undue risks.

2.6 Scientific Motivation

The scientific motivation of this study is to improve patient-level T1c supplementation at inference time. Recently, the BraSyn challenge explicitly targeted this scenario, aiming to substitute missing MRIs with synthetic scans to enable complete multi-modal segmentation inputs and avoid degraded performance (Jain et al., 2026; H. B. Li et al., 2024).

However, T1c synthesis remains comparatively underrepresented (Dayarathna et al., 2024; Dogra et al., 2026; Eidex et al., 2025). Gadolinium-enhancement patterns are highly heterogeneous and have no equivalent in non-contrast scans, making tumor anatomy learning challenging (Dong et al., 2026; Eidex et al., 2026; Tabassum et al., 2024). Models developed for some scans may struggle to generalize to others and are often not evaluated on T1c synthesis (Dayarathna et al., 2024; Ibrahim et al., 2025). Furthermore, synthetic brain MRI is often evaluated on image fidelity

alone, leaving downstream clinical utility (e.g., segmentation performance) unknown (Dayarathna et al., 2024; S. Wang et al., 2025; Wu et al., 2024). Since segmentation models rely on large, complete datasets, T1c missingness can meaningfully degrade performance (Kim & Park, 2024; Raut et al., 2024).

It therefore remains unknown to what degree state-of-the-art generative models used in other modalities can be adapted to synthesize high-fidelity T1c for downstream segmentation supplementation.

2.7 Research Strategy

The research aims of this study serve to address the societal and scientific motivations presented above by systematically evaluating synthetic T1c for both image fidelity and downstream segmentation, emphasizing patient-level supplementation at inference time.

Assessing synthetic image quality is central to evaluating to what degree generative models can produce realistic T1c scans. This motivates the first research question:

RQ1: To what extent can state-of-the-art generative models generate T1c images that are comparable in quality to ground-truth T1c?

This RQ evaluates how well state-of-the-art models — unconditional and conditional — can synthesize T1c, with conditioning strategy treated as a form of input feature selection. Image fidelity was measured across distributional, pixel-wise/structural, and perceptual dimensions.

Examining unconditional T1c distribution learnability establishes to what extent conditional T1c synthesis failures reflect the modality itself or model architecture and conditioning strategy. Therefore, this study explores the following sub-question:

SQ1: To what extent can an unconditional model learn the T1c distribution? Since unconditioned outputs cannot be compared to ground-truth scans, SQ1 was evaluated solely on distributional quality – specifically, how closely the synthetic image distribution matches the real T1c distribution.

Testing against an external dataset helps determine the extent to which models adapt to other image distributions. This motivates the second sub-question:

SQ2: To what extent can state-of-the-art models generalize to an external dataset? Conditional models generated images that were conditioned on an external dataset, with identical fidelity assessment to RQ1. StyleGAN2-ADA images were evaluated on the same external dataset identically to SQ1.

Synthetic T1c replacement in segmentation more directly assesses clinical utility than image fidelity metrics alone. Therefore, this study explores the second research question:

RQ2: To what extent can synthetic conditional T1c images substitute for missing T1c in downstream segmentation?

A segmentation model trained on complete, real data – approximating real-world clinical segmentation – was applied to five conditions using complete non-contrast scans and varied T1c. Three simulated realistic T1c availability (0%, 50%, or 100% missing T1c), and two reflected missing T1c substitution with conditioned scans (50% real and 50% synthetic, or 100% synthetic T1c). Segmentation performance was assessed using Dice and IoU scores.

Segmentation performance may vary by tumor size and region, with under-performance for specific tumor phenotypes disproportionately affecting patients relying on their accurate segmentation. Therefore, this study investigated the following sub-question:

SQ3: To what extent does synthetic T1c segmentation performance vary by tumor size/region? SQ3 examines the disparate impact across tumor size and subregion, identifying to what degree segmentation performance varies across them with synthetic T1c supplementation. RQ2 scores were therefore stratified by tumor size and subregion.

Testing segmentation performance on external data helps determine whether synthetic T1c substitution is more widely applicable across patient populations and acquisition contexts. This motivates the final sub-question:

SQ4: To what extent does synthetic T1c segmentation performance generalize to an external dataset? SQ4 seeks to address the segmentation generalizability gap by applying identical strategies to RQ2 for conditioned synthetic T1c on an external dataset.

2.8 Findings

SynDiff and pix2pixRAD generated high-fidelity T1c, though fidelity depended on conditioning strategy and model type. The custom diffusion model failed to synthesize high-fidelity T1c. StyleGAN2-ADA demonstrated strong unconditional T1c synthesis, suggesting the T1c distribution is learnable, though generalization was limited. Using synthetic T1c in place of missing T1c improved segmentation performance compared to missing T1c conditions but did not reach real data performance. Performance was lowest in small tumors and enhancing tumor regions, indicating disparate clinical impact for patients reliant on accurate assessment of these features.

3 LITERATURE REVIEW

3.1 State-of-the-Art Brain MRI Synthesis

3.1.1 GAN-based Synthesis

GAN use in brain MRI spans multiple tasks, including super-resolution and reconstruction (Tavse et al., 2022), with augmentation and translation being the most frequent (Ali et al., 2022; Dimitriadis et al., 2022; Ge et al., 2020).

GAN architectures for augmentation encompass both image-translation and unconditional models (Ali et al., 2022; Islam et al., 2026). Unconditional GANs are employed when the goal is single-modality learning without cross-modality mapping (Dimitriadis et al., 2022; Islam et al., 2026). Common modifications include deep-convolutional networks (DCGAN) or Wasserstein loss (WGAN), improving brain MRI image fidelity compared to baseline (vanilla) GANs (Cepa et al., 2024; Han et al., 2018; Mehewish et al., 2026; Tavse et al., 2022). Style-based GANs stabilize image feature learning through style vectors in individual generator layers (Karras et al., 2020). Considered state-of-the-art, StyleGANs have been found to outperform DCGAN and WGAN in image distribution fidelity on multiple medical imaging modalities (Skandarani et al., 2023). StyleGAN2-ADA, which uses adaptive discriminator augmentation to increase discriminator generalization, was used to synthesize high-quality healthy and glioma T1w scans (Lai et al., 2025; Tariq et al., 2023).

Image-to-image translation tasks typically fall into paired (pix2pix) or unpaired (cycleGAN) conditional approaches. Each uses different methods to learn mappings between brain MRI modalities: pix2pix models leverage adversarial and reconstruction (L1) losses to predict specific ground-truth

scans, making them better suited for synthesizing missing modalities using existing scans (Conte et al., 2021; Isola et al., 2017); while cycleGANs apply additional cycle consistency loss to generate novel images without paired images (J.-Y. Zhu et al., 2017).

Though conditional GANs traditionally use binary cross-entropy (BCE) adversarial loss, studies frequently substitute or augment this with auxiliary terms to stabilize training and improve modality-specific learning (Abod & Aziz, 2026; Baltruschat et al., 2024; Tavse et al., 2022). For example, replacing BCE with least-squares loss stabilized FLAIR and multi-modal translation (Chan et al., 2023; Touati & Kadoury, 2023), gradient loss was applied to high-fidelity multi-modal synthesis (Dai et al., 2020; Naseem et al., 2022), and perceptual loss outperformed other auxiliary losses in T1w and T2w synthesis (Baltruschat et al., 2024).

Pix2pix architectures are frequently used for missing modality synthesis. For instance, a standard pix2pix was successfully employed in 2D mappings from T2w to FLAIR and T1c to T1w (Conte et al., 2021; Zoghby et al., 2024), multi-input 2D mappings to T1c (L. Li et al., 2024), and 3D glioma T1c synthesis (Abod & Aziz, 2026; Raut et al., 2024).

Conditional GANs face several limitations, including training instability and mode collapse, where the generator learns to produce limited outputs (Goodfellow et al., 2014; Saad et al., 2024). Pix2pix's reconstruction loss encourages pixel-wise correspondence to ground truth, potentially pushing the model towards blurred outputs (Isola et al., 2017). Finally, adversarial training may struggle to capture complex distributions without adjustments to loss functions or architecture (Saad et al., 2024; Skandarani et al., 2023).

3.1.2 *Diffusion Model-Based Synthesis*

Diffusion models (DDPMs) have emerged as a strong alternative to GANs in brain MRI, with similar applications including super-resolution (Y. Mao et al., 2023), modality conversion (Lyu & Wang, 2022), data augmentation and image-to-image translation (Kazerouni et al., 2023; Özbey et al., 2023).

Like GANs, diffusion models can be roughly sorted into unconditional and conditional approaches for augmentation and missing modality synthesis, respectively (Friedrich et al., 2025; Kazerouni et al., 2023; Nafi et al., 2024). Unconditional diffusion models have been used for 2D brain MRI tumor insertion (Rouzrokh et al., 2023), multimodal 2D synthesis (Usman Akbar et al., 2024), as well as high-fidelity 3D healthy and Parkinson's T1w synthesis (Mallardi et al., 2025).

Conditional DDPMs have been applied to 2D translation tasks, including T1w to T2w synthesis (Kiani Kalejahi et al., 2026), and have outperformed GANs in image fidelity for glioma 2D FLAIR (Mirowski & Fabijańska, 2026) and glioma 3D multi-modal synthesis (Dorjsembe et al.,

2024) when conditioning using tumor segmentation masks. 3D applications of DDPM have also been used in synthesis of randomly missing brain MRI modalities, including T1c (Meng et al., 2024).

Despite their advantages, diffusion models are often slow and computationally expensive (Kazerouni et al., 2023; Nafi et al., 2024), and may be prone to memorization of training images, especially on small datasets (Akbar et al., 2024). Unlike GANs, DDPMs typically rely on mean-squared error noise-prediction loss, with efficiency improvements principally achieved through architectural or sampling changes (Ho et al., 2020; Kumar et al., 2025; Rombach et al., 2022). Notably, SynDiff (Özbey et al., 2023) reduces computational demands compared to standard DDPMs by combining adversarial and diffusion components in a hybrid architecture for multimodal 2D translation. Using a GAN module to provide an initial image estimate that guides diffusion, it decreases the number of diffusion steps and achieves higher-fidelity synthesis across multiple non-contrast MRI modalities than GAN and DDPM approaches.

Further architectural improvements include the latent diffusion model (LDM), which reduces computational cost by compressing images into a latent representation before diffusion (Rombach et al., 2022). Brain MRI applications include unconditioned synthesis of high-fidelity 3D healthy T1w (Pinaya et al., 2023) and 3D T1c (Kim & Park, 2024), as well as conditional glioma 3D multi-modal T1c synthesis (Eidex et al., 2026). DDIMs (denoising diffusion implicit models) improve generation speed through sampling reduction in place of strict stepwise denoising (Song et al., 2021). However, to the author’s knowledge, DDIM in brain MRI synthesis remains limited to 2D synthesis from a brain tumor classification dataset (Peng et al., 2024) and combined use with LDM in unconditioned 3D healthy T1w generation (Pinaya et al., 2023).

3.2 *Synthetic Images in Downstream Segmentation Tasks*

Beyond image fidelity, studies have assessed synthetic scan utility for downstream tasks, including dataset augmentation for tumor segmentation. Ferreira et al. (2024) achieved state-of-the-art segmentation performance in the BraTS 2023 Glioma Segmentation challenge using a conditional paired GAN (GliGAN) to insert tumors into healthy scans, with Jain et al. (2026) winning the 2025 challenge by integrating GliGANs directly into segmentation training. Usman Akbar et al. (2024) demonstrated segmentation training data augmented with StyleGAN and unconditional diffusion synthetic images could reach 80-90% of real image performance. Additionally, a 3D segmentation model trained on mixed real and 3D-

DDPM model synthetic multimodal scans (including T1c) outperformed one trained on real data alone (Dorjsembe et al., 2024).

Research has also addressed scan missingness by testing substitution of missing MRI modalities with synthetic scans at inference time. Replacing 2D T1w and FLAIR with pix2pix-generated versions achieved similar segmentation performance to complete scans (Conte et al., 2021; Zoghby et al., 2024). Replacement with high-fidelity 3D T1c, T1w, and FLAIR from a conditional GAN with segmentation loss achieved slightly better segmentation performance than with real scans alone (Abod & Aziz, 2026), while a pix2pixNifTI model achieved strong multimodal 3D substitution performance (Raut et al., 2024). Finally, GANs with tailored loss functions demonstrated state-of-the-art segmentation scores across multiple replaced modalities (Baltruschat et al., 2024; Carpentiero et al., 2026; H. B. Li et al., 2024).

3.2.1 Literature Gaps

Brain MRI synthesis studies commonly evaluate image quality along a single dimension, painting a potentially misleading picture of fidelity (Deo et al., 2025; Dohmen et al., 2025). For instance, pixel-wise and structural fidelity are widely used despite difficulty capturing image blurriness and poorly predicting perceptual distortions (Deo et al., 2025; Dohmen et al., 2025). Furthermore, studies frequently rely on image fidelity alone to assess image quality, leaving downstream clinical utility untested (Dayarathna et al., 2024; S. Wang et al., 2025; Wu et al., 2024). When segmentation is performed, total performance may be reported without tumor subregion analysis (Abod & Aziz, 2026; Baltruschat et al., 2024; Dorjsembe et al., 2024).

Synthetic image segmentation is frequently conducted using GAN-based methods, with diffusion-based studies only recently emerging (Azad et al., 2026; Dayarathna et al., 2024; Eidex et al., 2026). Additionally, image fidelity and segmentation are frequently not assessed for generalization on external datasets (Dayarathna et al., 2024; Ibrahim et al., 2025; S. Wang et al., 2025).

T1c generation is underrepresented compared to non-contrast modalities (Dayarathna et al., 2024; Dogra et al., 2026; Eidex et al., 2025), with synthetic T1c in downstream segmentation also understudied (Dayarathna et al., 2024; Moya-Sáez et al., 2023). T1c substitution tends to yield the poorest segmentation performance compared to other modalities, especially in contrast-dependent subregions (Raut et al., 2024). Even otherwise high-performing T1c segmentation has yielded low segmentation scores in contrast-dependent regions (Huang et al., 2022).

This study addresses these gaps by adapting representative state-of-the-art 2D generative models developed for non-contrast brain MRI synthesis to T1c for the first time. The representative conditional GAN is pix2pixRAD (Conte et al., 2021) – a vanilla pix2pix – allowing assessment of both baseline performance and the effect of modifications to inputs and loss functions on T1c synthesis. The representative conditional diffusion models are SynDiff (Özbey et al., 2023), evaluated across multiple conditioning strategies, and a segmentation-mask conditioned DDPM examining mask-only T1c conditioning (Mirowski & Fabijańska, 2026). Finally, StyleGAN2-ADA serves as the unconditional representative, assessing T1c learnability independent of conditioning (Karras et al., 2020).

4 METHODOLOGY

4.1 Datasets

4.1.1 BraTS-2023 Adult Glioma Dataset

This study primarily used the BraTS-2023 Adult Glioma dataset, an open-access dataset of pre-treatment adult glioma MRI scans (Baid et al., 2021; Bakas et al., 2017a, 2017b, 2017c; Menze et al., 2015). MRIs were collected using various scanners and protocols at 19 different institutions located across Germany, Hungary, India, Switzerland, and the United States (Bonato et al., 2025). Scanner models and associated hardware are not enumerated, and tumor type/grade (e.g., HGG) were not made available.

Only the provided training set was used, as the validation set lacked segmentation masks. The training set consisted of 1,251 complete scans, each scan folder containing T1-weighted pre-contrast (T1w), T1-weighted post-contrast (T1c), T2-weighted (T2w), T2-weighted FLAIR scans, and corresponding segmentation masks. All brain scan images were preprocessed (e.g., skull-stripped). Segmentation mask label values correspond to pixel values representing the following sub-regions: enhancing tumor (ET), sites of blood-brain barrier disruption from tumor growth (value: 3); Peritumoral edema (ED), or infiltrated tissue (value: 2); and necrotic tumor core (NCR), dead/non-functioning tissue at tumor center (value: 1). All other pixels were 0.

4.1.2 UCSF-PDGM Dataset

To evaluate generalizability this study employed the University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM) dataset (Calabrese et al. 2023; *hereafter UCSF*). This dataset includes preoperative

brain scans of 501 glioma patients (6 follow-up). Data was collected between 2015–2021 using a single scanner type at one institution. Tumor types consisted of 55 patients (11%) of grade II, 42 patients (9%) of grade III, and 403 patients (80%) of grade IV tumor diagnosis.

4.1.3 Preprocessing

Native BraTS scans were NIfTI files of dimension (240 x 240 x 155). 155 2D axial .png slices were generated per scan, then resized to (256 x 256) with pixel intensity range [0, 255]. Slices containing $\leq 5\%$ brain matter were removed to ensure sufficient anatomical content, and 15 of the remaining slices were selected at evenly spaced intervals to include clinically relevant brain regions (Mirowski & Fabijańska, 2026). This was done due to computational resource constraints. While 15 slices per patient were used for all model training, only the best-performing SynDiff configuration was subsequently retrained from scratch on 100 slices per patient to replicate the authors' original data conditions (Özbey et al., 2023), with preprocessing identical to that described above.

Data were split at the scan level into train and test (1000 : 251) sets (Raut et al., 2024). However, a small, hold-out validation set was created post-hoc from 10% of the training set to accommodate hard-coded training specifications (train = 900: validation = 100, test = 251). Only StyleGAN2-ADA was trained on the full training set (80%). Data were further preprocessed according to the specific needs of each model.

Pix2pixRAD data were kept at (256 x 256) and transformed to 3-channel RGB to match expected input, with pixel intensities normalized to [-1,1] and additional random cropping and horizontal flipping augmentation. StyleGAN2-ADA data were resized to (512 x 512) to accommodate pre-trained weights and normalized to [-1,1]. SynDiff inputs were kept at the original resolution, normalized to [-1,1] and stored in .mat files. Finally, the custom diffusion model data were center cropped (160 x 224) with T1c targets normalized to [0,1]. UCSF scans were processed into a single, unsplit dataset (495 patients, and 6 follow-up) with identical preprocessing to BraTS. See Figure 1 for a visual representation of preprocessing.

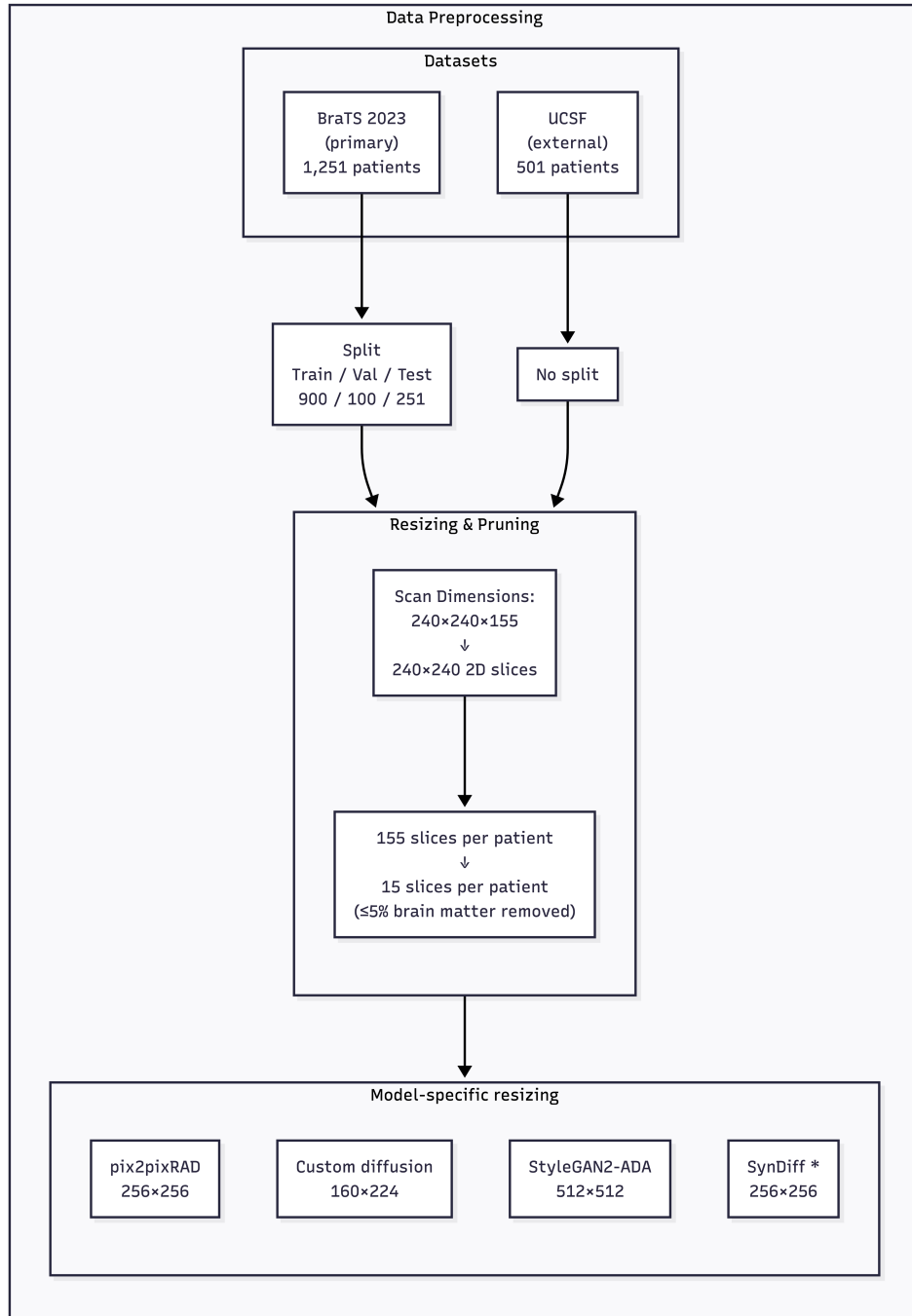


Figure 1: Preprocessing pipeline for BraTS and UCSF datasets. * The best model configuration for SynDiff was trained on 100 axial slices per patient, which was pruned in the same manner.

Scaled segmentation mask pixel values were scaled to $[0,255]$, and modified segmentation masks were created by including healthy brain tissue at a pixel value of 4 and scaling upwards to $[0,255]$ (Figure 2).

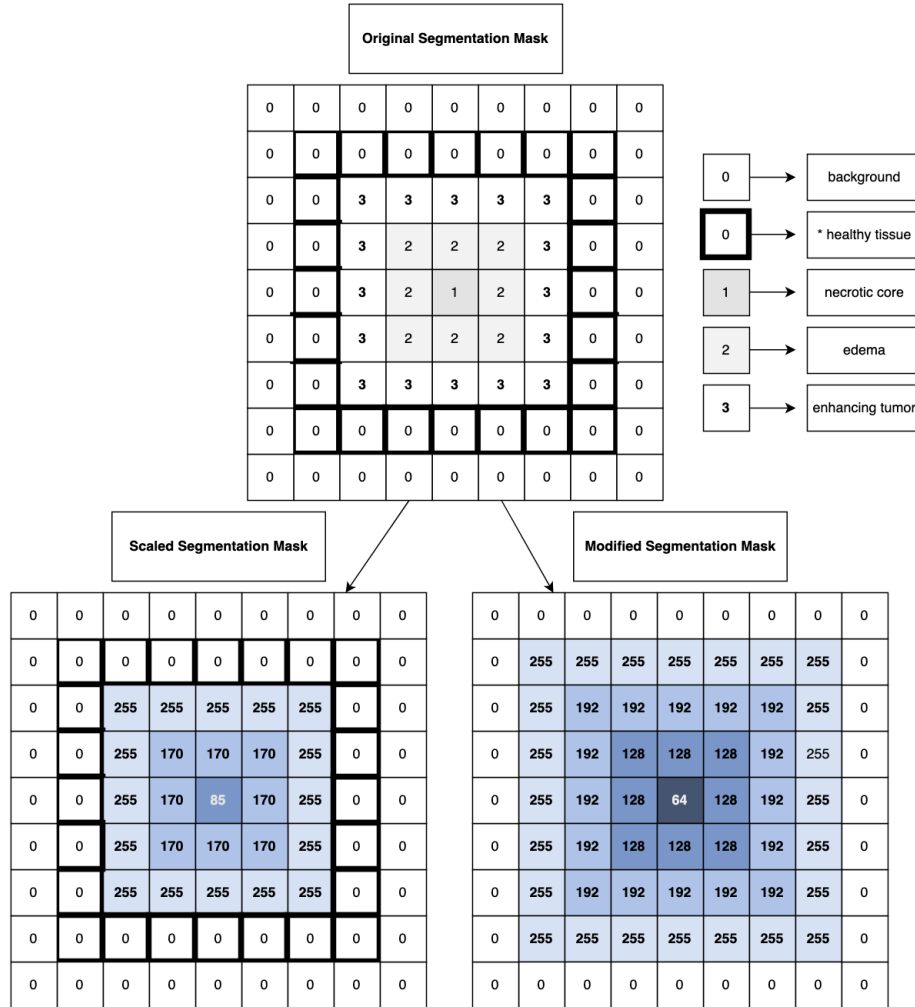


Figure 2: This figure depicts segmentation mask transformations. Tumor regions are denoted by pixel values 1-3, while 0 denotes background (* healthy brain tissue is shown as 0 with bold borders for additional clarity). Modified segmentation masks were made by setting healthy brain tissue to 4, then scaling to range $[0,255]$. Scaled segmentation masks applied the same scaling, but did not include healthy brain tissue.

4.1.4 Dataset Comparability: Error Analysis

Signal-to-Noise Ratio (SNR) and Contrast-to-Noise Ratio (CNR) were calculated per scan type per dataset to measure overall image quality and tissue

differentiation, respectively (Magnotta & Friedman, 2006). Wasserstein’s W_1 was used to compare dataset distributions, as it does not assume normality (Villani, 2009). Significant differences between datasets were gauged using the two-tailed Mann-Whitney U test.

4.2 *T1c Synthesis (RQ_1 , SQ_1 , SQ_2)*

Four models were applied to $T1c$ synthesis: two GANs (pix2pixRAD, StyleGAN2-ADA) and two diffusion models (SynDiff, custom diffusion). Major architectural changes and 3D synthesis were considered out-of-scope – the former to avoid confounds in conditioning strategy and input configuration assessment, and the latter due to computational cost. Detailed training and model descriptions are available in Appendices 6.11–6.14.

4.2.1 *Pix2pixRAD*

Pix2pixRAD (Conte et al., 2021) – a vanilla pix2pix identical to Isola et al. (2017) – was selected as the conditional GAN representative, given its demonstrated $T1c \rightarrow T1w$ synthesis and comparable segmentation performance to genuine scans. Its flexibility allows adaptation to multiple inputs (L. Li et al., 2024) and loss functions (Abod & Aziz, 2026; Baltruschat et al., 2024), supporting targeted modifications. Additionally, a vanilla baseline avoids confounds, allowing modifications to be attributed to $T1c$ synthesis effects. CycleGAN architectures were not considered because unpaired translation does not suit missing-modality replacement.

Initial training employed baseline hyperparameter settings of Conte et al. (2021), described in Appendix A1: Pix2pixRAD. Least-squares loss (LSGAN) replaced BCE to prevent gradient saturation with discriminator loss (X. Mao et al., 2017). Perceptual (VGG-16) and gradient losses were added to encourage image sharpness and anatomical region differentiation (Johnson et al., 2016; Ma et al., 2020). Cityscape dataset pretrained weights were employed to promote structural learning (Cordts et al., 2016). The final input configuration consisted of $T1w$, modified segmentation mask, and FLAIR – one per input channel. $T1w$ was used for its structural similarity to $T1c$ and demonstrated high-fidelity $T1c$ to $T1w$ synthesis (Conte et al., 2021), FLAIR for additional anatomical context, and a modified segmentation mask (Figure 2) for tumor spatial information (see Figure 3 for an input channel sample). Full training description is reported in Appendix A1: Pix2pixRAD.

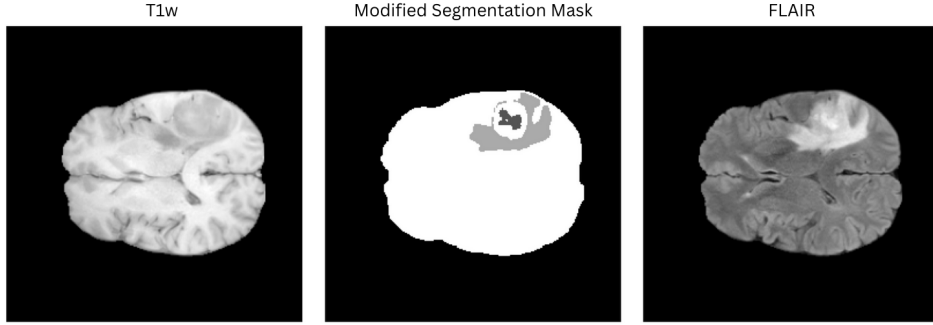


Figure 3: Sample of a 3-Channel input for pix2pixRAD, with T1w in the first channel, the modified segmentation mask (Figure 2) in the second, and FLAIR in the third.

4.2.2 SynDiff

SynDiff (Özbey et al., 2023) was selected as the conditional diffusion representative for state-of-the-art cross-modality translation. Its hybrid adversarial-diffusion design accommodates both paired and unpaired training data, allowing flexible modality translation, and it demonstrated higher-fidelity synthesis across multiple non-contrast MRI modalities compared to numerous GAN and DDPM architectures (Özbey et al., 2023). However, its performance on T1c synthesis, and downstream segmentation, remains untested.

While LDM has been applied to T1c synthesis (Eidex et al., 2026; Kim & Park, 2024) and brain MRI more broadly (Pinaya et al., 2023; L. Zhu et al., 2023), these were done for 3D synthesis, rendering it out-of-scope. DDIM, as an alternate sampling strategy, could have been applied to SynDiff; however, this introduces sampling strategy as an additional variable, making it difficult to attribute performance differences to input configurations alone, rendering DDIM out-of-scope.

Baseline SynDiff is not multi-input, therefore four single input configurations were tested: FLAIR, T1w, and two modified T1w scans. FLAIR was tested as an analogous mapping to $T2w \rightarrow T1w$ (Özbey et al., 2023), T1w due to its best performance for conditioned 3D T1c synthesis (Kim & Park, 2024), and T1w modified with tumor priors to assess tumor information guidance. These were created by concatenating T1w with a scaled segmentation mask (Figure 2) such that: $\text{modified T1w} = T1w + \alpha \times \text{scaled segmentation mask}$, using $\alpha = 0.1$ and $\alpha = 0.3$. All four configurations were trained on 15 axial slices per patient. The best-performing configuration (modified T1w, $\alpha = 0.3$) was retrained from scratch on 100 axial slices per patient to replicate Özbey et al. (2023) conditions. See Appendix A2: SynDiff for architectural and training details.

4.2.3 Custom Diffusion

The custom diffusion model (Mirowski & Fabijańska, 2026) was selected for its distinct conditioning strategy: generating high-fidelity FLAIR images with high segmentation performance conditioned only on modified segmentation masks (Figure 2). A tumor-conditioned 3D-DDPM similarly synthesized multiple modalities including T1c from segmentation masks, approaching real image segmentation performance (Dorjsembe et al., 2024). Therefore, this study extends evaluation to 2D T1c conditioning with spatial tumor priors alone.

The baseline configuration showed least fidelity degradation across hyperparameter configurations, and was selected for segmentation evaluation. Training and architectural details are available in Appendix A3: Custom Diffusion.

4.2.4 StyleGAN2-ADA

StyleGAN2-ADA (Karras et al., 2020) was included as the state-of-the-art unconditional model to test T1c distribution learnability independent of conditioning.

Style-based GANs outperformed earlier unconditional variants like DCGAN and WGAN in image distribution fidelity across medical imaging modalities (Skandarani et al., 2023), as well as an unconditional DDPM in multi-modal brain MRI (Usman Akbar et al., 2024). StyleGAN2-ADA has demonstrated high-quality glioma T1w (Tariq et al., 2023) and healthy T1w synthesis (Lai et al., 2025), and its successful employment of non-medical domain weights suggests strong domain flexibility (Tariq et al., 2023). However, its performance on T1c synthesis has not been evaluated.

An unconditional diffusion model was not included alongside StyleGAN2-ADA, as SQ1 seeks to assess T1c distribution learnability, rather than benchmark unconditional architectures. Furthermore, existing DDPM approaches have been narrower in scope, like tumor inpainting (Rouzrokh et al., 2023) and healthy/Parkinson's T1w generation (Mallardi et al., 2025).

Pretrained AFHQ wild (512 x 512) weights (Choi et al., 2020) were implemented, following Tariq et al. (2023). The generator at 1,880 king (thousands of images seen during training) was selected based on highest distributional fidelity (see Appendix A4: StyleGAN2-ADA).

4.2.5 Image Fidelity Metrics

Synthetic brain MRI quality can be assessed along three dimensions: distributional, pixel-wise/structural, and perceptual fidelity, with a combined approach allowing for a more holistic fidelity assessment (Deo et al., 2025; Dohmen et al., 2025).

Distributional fidelity in all models is measured by the Fréchet Inception Distance (FID), quantifying distribution-level distance, and Kernel Inception Distance (KID), reporting the mean distribution distance between individual images (Bińkowski et al., 2021; Heusel et al., 2017).

Conditional models were assessed for pixel-wise similarity using Peak Signal-to-Noise Ratio (PSNR), and structural anatomical learning using Structural Similarity Index Measure (SSIM) (Z. Wang et al., 2004). Perceptual fidelity was measured with Gradient Magnitude Similarity Deviation (GMSD), which captures local edge-level differences (Xue et al., 2014); and Perceptual Image Patch Similarity (LPIPS), which estimates visual similarity as experienced by human observers (Zhang et al., 2018). These metrics were not calculated for unconditioned images, as they require predicted image versus ground truth comparisons.

GMSD, LPIPS, and SSIM values range from $[0,1]$, with higher SSIM indicating better quality, and the reverse for the others. PSNR is measured in decibels (dB), with higher values representing closer pixel-wise correspondence. KID and FID values do not have an upper bound, but lower FID values and KID values approaching 0 indicate high distributional similarity (Dohmen et al., 2025).

4.3 Segmentation (RQ2, SQ3, SQ4)

4.3.1 Segmentation Model

Synthetic T1c substitution was evaluated using the nnU-Net 2D segmentation model, given its strong performance in brain tumor segmentation and self-configuring architecture (Isensee et al., 2021; H. M. Luu & Park, 2021). The nnU-Net was trained on full, real, unpruned training data (155 slices per patient; 1000 patients). Final model weights were obtained through 5-fold cross validation.

4.3.2 Segmentation Metrics

Tumor segmentation performance was gauged with the Dice Similarity Coefficient (Dice score), and Intersection over Union (IoU) score, both measuring the degree to which two datasets overlap. Scores range from $(0,1)$, with 1 indicating perfect overlap. Segmentation failure patterns between datasets were investigated using false positive (FP), false negative (FN), and complete miss rates (Dice = 0), calculated per tumor region and size.

$$\text{Dice}(A, B) = \frac{2|A \cap B|}{|A| + |B|} \quad \text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

4.3.3 Segmentation Evaluation

Segmentation was conducted over three composite tumor regions: Enhancing Tumor (ET), representing active tumor tissue; Tumor Core (TC), corresponding to ET and the necrotic core (NCR); and Whole Tumor (WT), consisting of all tumor subregions.

Predicted segmentation masks were generated from each condition and evaluated against ground-truth masks. Metrics were obtained per patient and region, then averaged across both. Results were further stratified by tumor size using bins derived from total tumor pixels per patient from the cross-validated nnU-Net. Tumors were binned by percentiles: small ($<Q_{33}$), medium ($Q_{33} \leq Q_{66}$), and large ($>Q_{66}$) (Table 1). UCSF dataset segmentation results were stratified by tumor region but not tumor size due to time constraints.

Table 1: Tumor size thresholds (in pixels) used to stratify patients into small, medium, and large categories for each subregion. Thresholds are defined as the 33rd and 66th percentiles of the training set pixel counts; the large category contains all cases at or above the 66th percentile.

Size	Pixels		
	<i>NCR</i>	<i>Edema</i>	<i>ET</i>
Small	254.95	2576.35	646.34
Medium	1285.48	6770.10	2276.84
Large	4955.91	13447.38	4895.94

4.4 Implementation Details

Training, segmentation, and analyses were conducted on NVIDIA A40 GPUs (46GB VRAM, Driver 535.261.03, CUDA 12.2) provided by Tilburg University’s GPU4EDU program. Models were implemented in dedicated Conda environments (dependencies shown in Appendix A, Table 11).

Following training, all models generated 3,765 T1c scans (15 scans per patient; 251 patients) for evaluation against the BraTS test set. For generalization assessment, conditional models generated 7,515 T1c scans (15 scans per patient; 501 patients) conditioned on UCSF, with StyleGAN2-ADA generating an equivalent number of unconditioned scans (See Figure 4 for evaluation workflow).

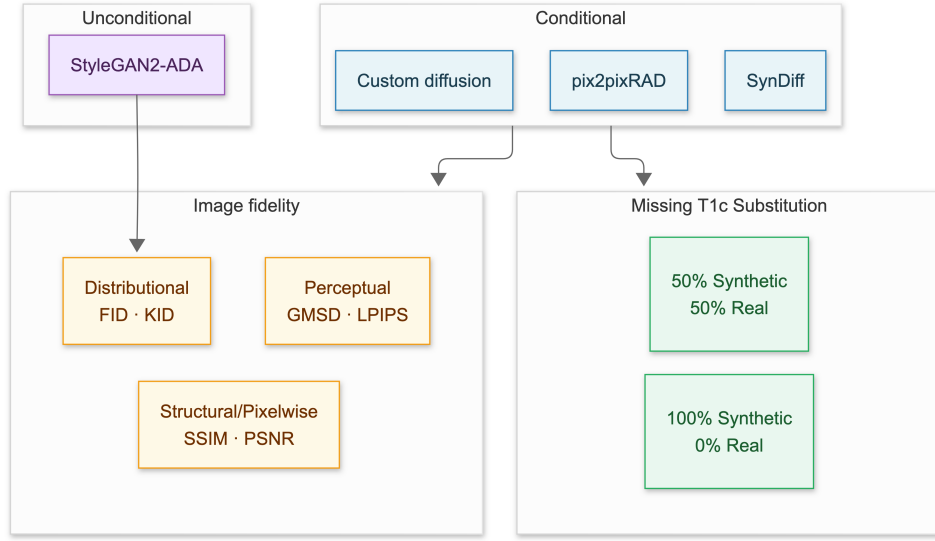


Figure 4: Synthetic image evaluation pipeline. Unconditional images are evaluated for distributional fidelity against BraTS test data only. Conditional images are evaluated on all three dimensions of image fidelity, and are evaluated as T1c substitutes in downstream segmentation. Conditional models generated synthetic T1c images conditioned on the UCSF dataset to evaluate generalizability. These models underwent the same fidelity and segmentation performance evaluation.

Synthetic condition datasets consisted of the BraTS test set containing real non-contrast (T1w, T2w, and FLAIR) scans, and T1c scans that were either 100% synthetic, or 50%-real, 50%-synthetic. The 50%-synthetic split was done at the patient level.

Synthetic condition performance was compared to the cross-validated nn-UNet baseline (trained on the complete, real BraTS train dataset) and to T1c missingness conditions. These were composed of the BraTS test dataset containing non-contrast scans, and 0%, 50%, or 100% missing T1c (Figure 5). Missing T1c was simulated with images of uniform pixel value 0.

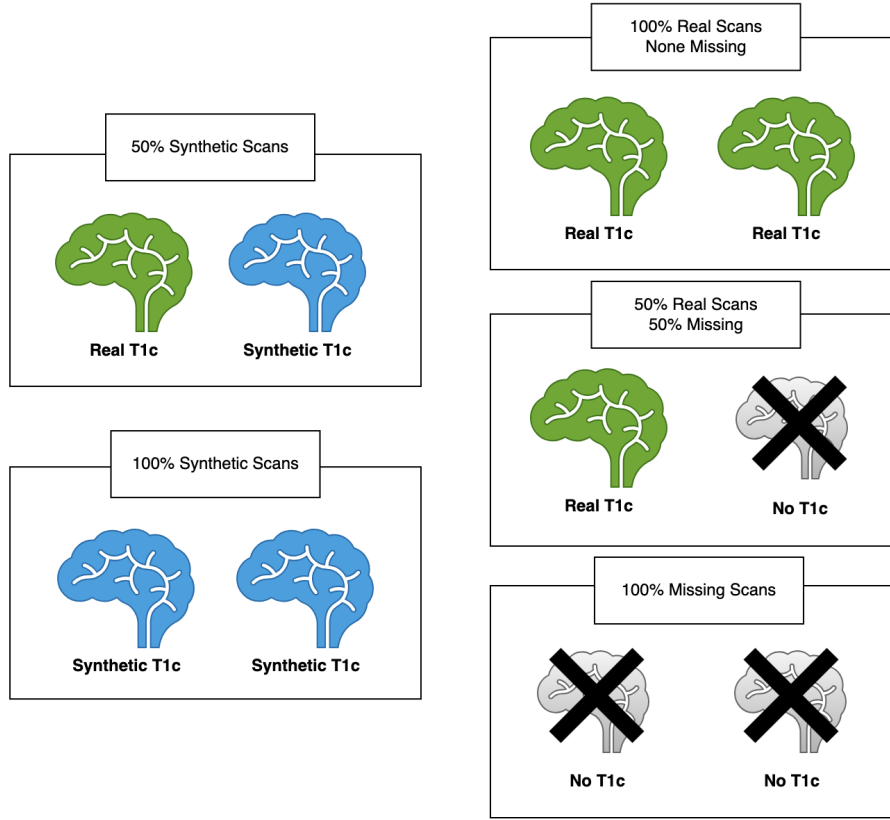


Figure 5: This image depicts the BraTS test set conditions for segmentation evaluation. The left conditions denote the synthetic T1c replacement conditions, where conditioned images replace either 50% or 100% of T1c in the BraTS test set. These are compared in segmentation performance to the conditions presented on the right, which represent real-world dataset missingness scenarios: the BraTS test data with 0% missing T1c, 50% missing T1c, and 100% missing T1c. These same conditions are applied to the UCSF dataset to examine generalizability of synthetic T1c as a replacement for missing T1c.

5 RESULTS

5.1 Image Fidelity

Final configurations were selected based on best fidelity performance after training (see Appendices 6.11–6.14).

Sample final generator predictions and select intermediate training samples are shown for pix2pixRAD (Figure 6) and SynDiff (Figure 7).

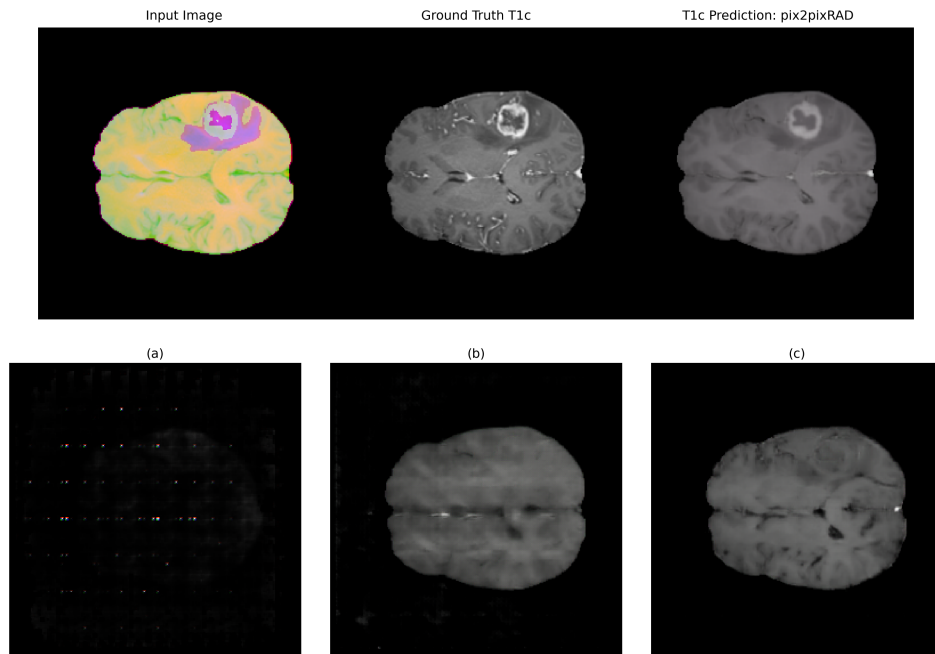


Figure 6: Examples of pix2pixRAD model outputs at various stages of training. The top right of the figure is the prediction of the best performing generator. The image on the top left is the input, consisting of a 3-channel image with T1w, a modified segmentation mask, and FLAIR. The top-center image is the ground truth T1c. Images labeled (a), (b), and (c) represent images generated using default hyperparameters and T1w input, adjusted hyperparameters with additional segmentation input, and final input configuration without LSGAN, respectively. Training details are available in Appendix A1: Pix2pixRAD.

Due to T1c synthesis failure in custom diffusion, intermediate training samples are found in Appendix A3: Custom Diffusion, Figure 19.

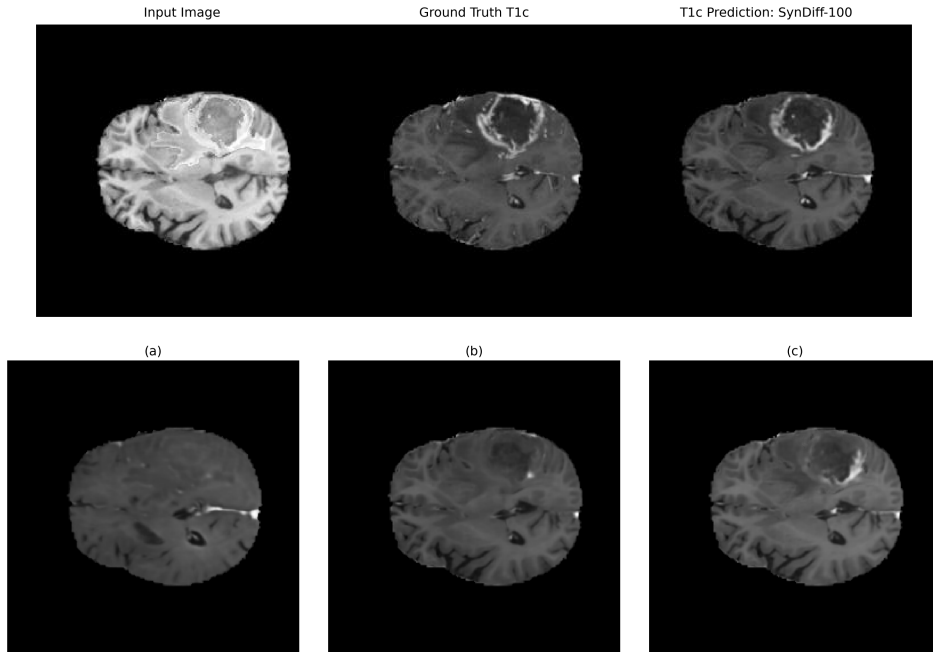


Figure 7: Examples of SynDiff model outputs at various stages of training. The top right of the figure (SynDiff-100) is the prediction of the best performing generator after training on 100 axial slices. The image on the top left is the input, consisting of T1w with a scaled segmentation overlay of intensity 0.3. The top-center image is the ground truth T1c. Images labeled (a), (b), and (c) represent images generated after training with 15 axial slices per patient using conditioned FLAIR, T1w, and T1w with a 0.1 intensity segmentation overlay, respectively. An example generated image of T1w with 0.3 segmentation overlay intensity after training for 15 axial slices is not depicted. Training details are available in Appendix A2: SynDiff.

While all models were trained on 15 slices per patient, the best-performing SynDiff configuration (modified T1w, $\alpha = 0.3$) was retrained from scratch on 100 slices per patient and used in all further analyses. Table 2 reports fidelity scores across SynDiff configurations.

Table 2: SynDiff fidelity metrics across input conditioning strategies.

Input Condition	α	Training Slices	FID	KID	SSIM	PSNR	GMSD	LPIPS
FLAIR	-	15	32.33	0.0205	0.9242	24.33	0.1453	0.0888
T1w	-	15	31.71	0.0218	0.9374	24.84	0.1298	0.0756
T1w + $\alpha \cdot$ Seg	0.1	15	28.72	0.0195	0.9372	25.36	0.1298	0.0736
T1w + $\alpha \cdot$ Seg	0.3	15	26.29	0.0176	0.9383	25.51	0.1267	0.0697
T1w + $\alpha \cdot$ Seg	0.3	100	18.52	0.0105	0.9415	25.90	0.1219	0.0592

Note: α controls the intensity of the scaled segmentation mask that was superimposed onto T1w. Training slices refers to the number of axial slices per patient that the model was trained with. All models save the bottom-most model were trained on 15 axial slices per patient.

StyleGAN2-ADA sample final generator synthetic images are shown in Figure 8. Image fidelity results for all final model configurations are reported in the following section.

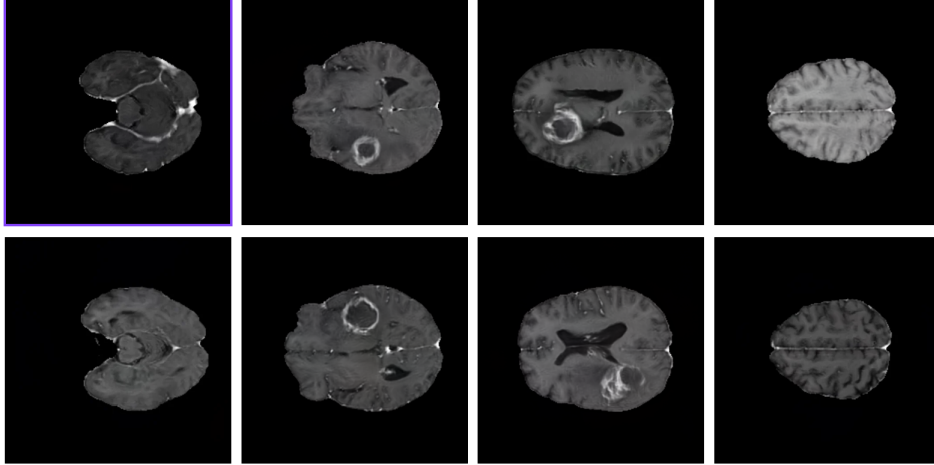


Figure 8: Synthetic images from the best-performing StyleGAN2-ADA generator.

5.1.1 Cross-Model Comparison: RQ1, SQ1, and SQ2

Table 3 and 4 refer to the cross-model comparisons on the BraTS test set and generalization dataset (UCSF). StyleGAN2-ADA was solely evaluated on FID and KID, as outputs are unconditioned. Final SynDiff (modified T1w, $\alpha = 0.3$; 100-slice training) was used in all comparisons.

SynDiff outperformed pix2pixRAD on image distribution metrics (FID: 18.52 vs. 37.17; KID: 0.0105 vs. 0.0260) on the BraTS test and UCSF datasets (FID: 19.35 vs. 48.83; KID: 0.0122 vs. 0.0339).

Table 3: Cross-model comparison of synthetic image fidelity using best model configurations (BraTS test-set).

Model	FID	KID	SSIM	PSNR	GMSD	LPIPS
Pix2pixRAD	37.17	0.0260	0.9509	27.62	0.1171	0.0634
SynDiff	18.52	0.0105	0.9415	25.90	0.1219	0.0592
Custom Diffusion	97.51	0.0843	0.1061	11.70	0.3482	0.4324
StyleGAN2-ADA	20.44	0.0060	—	—	—	—

Note: Cross model comparison of the best performing model configurations across image fidelity metrics for the BraTS dataset.

SynDiff’s image distribution scores decreased slightly with generalization, while pix2pixRAD’s scores decreased notably more (FID: 37.17 $\rightarrow$ 48.83; KID 0.0260 $\rightarrow$ 0.0339).

StyleGAN2-ADA achieved the second lowest FID score (20.44) and lowest KID score (0.0060). However, its distributional fidelity decreased on UCSF (FID: 68.75; KID: 0.0444).

Table 4: Cross-model comparison of synthetic image fidelity using best model configurations, generalized to the UCSF dataset

Model	FID	KID	SSIM	PSNR	GMSD	LPIPS
Pix2pixRAD	48.83	0.0339	0.9409	26.26	0.1164	0.0791
SynDiff	19.35	0.0122	0.9442	27.61	0.1198	0.0638
Custom Diffusion	67.66	0.0573	0.1608	15.57	0.2592	0.2572
StyleGAN2-ADA	68.75	0.0444	—	—	—	—

Note: Cross model comparison of the best performing model configurations across image fidelity metrics for the UCSF dataset.

Pix2pixRAD outperformed SynDiff on both SSIM (0.9509 vs. 0.9415) and PSNR (27.62 vs. 25.90). However, this reversed with UCSF, where SynDiff achieved better scores than pix2pixRAD (SSIM: 0.9442 vs. 0.9409; PSNR: 27.61 vs. 26.26).

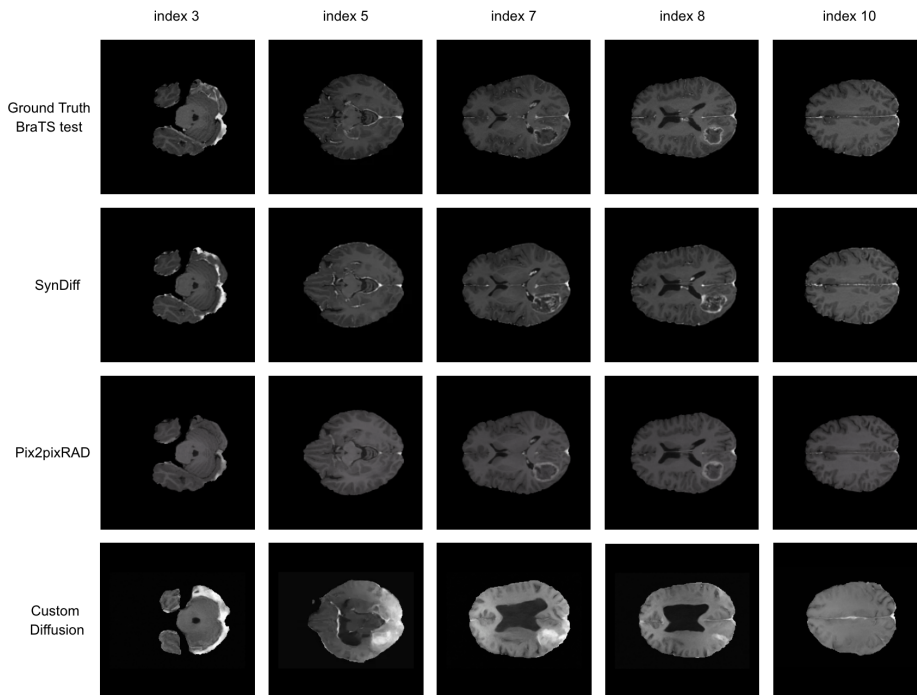


Figure 9: Sample axial slices of BraTS test set ground-truth T1c, and their corresponding predictions from the three conditioned models. The columns represent the index of the slice within one patient scan (full index ranged 0-14). The rows represent the source of the images. Slices were chosen to represent glioma tumors as well as healthy scans.

Finally, SynDiff achieved lower LPIPS scores than pix2pixRAD on both BraTS (0.0592 vs. 0.0634) and UCSF (0.0638 vs. 0.0791). Pix2pixRAD initially achieved a lower GMSD score than SynDiff on BraTS (0.1171 vs. 0.1219), but was outpaced by SynDiff in UCSF (0.1198 vs. 0.1164).

Custom diffusion performed the worst across all metrics for both datasets but did improve for each metric after generalization. Figures 9 and 10 depict final model predictions compared to ground-truth scans of BraTS and UCSF.

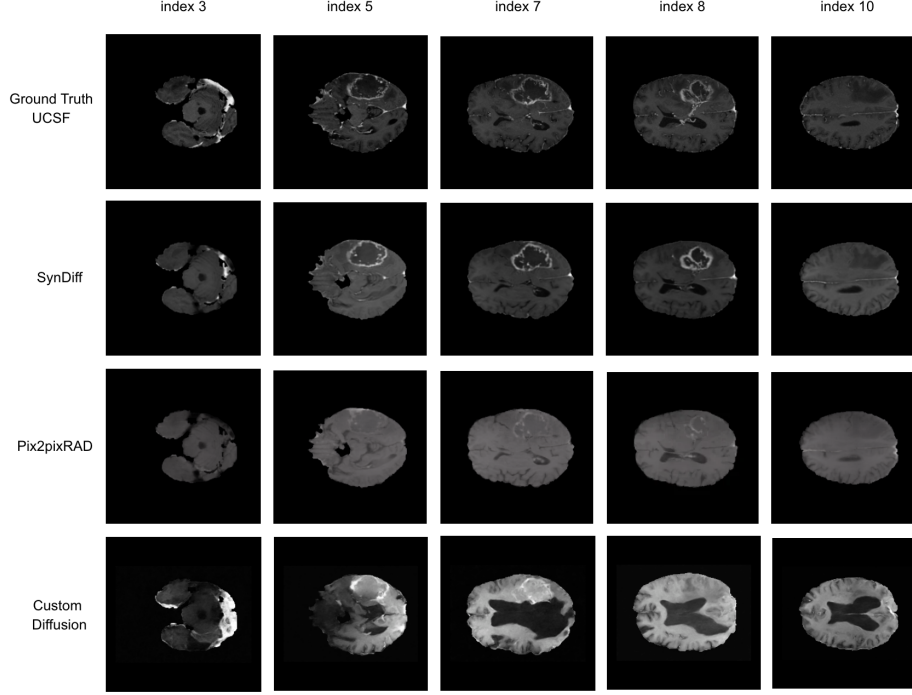


Figure 10: Sample axial slices of UCSF test set ground truth T1c, and their corresponding predictions from the three conditioned models. The columns represent the index of the slice within one patient scan (full index ranged 0-14). The rows represent the source of the images. Slices were chosen to represent glioma tumors as well as healthy scans.

5.2 Segmentation

5.2.1 Real Data Performance: BraTS Test Set and UCSF

Dice and IoU scores for direct BraTS test and UCSF dataset comparisons are reported in Table 5. All comparisons refer to Dice scores alone for brevity. Baseline nnU-Net performance on the complete training set after 5-fold cross validation (CV) yielded a mean Dice Score of 0.8599, and the following subregion Dice scores: 0.8261 (TC), 0.8685 (WT), and 0.8852 (ET). BraTS test performance degraded on the 0% missing condition (mean Dice = 0.7357), with subregion scores of 0.6754 (TC), 0.7490 (WT), and 0.7826 (ET). The region with the greatest performance decrease was TC, followed by WT, and ET.

However, UCSF segmentation performance for 0% missing T1c closely matched CV: TC (0.840), WT (0.833), and ET (0.845). Differences in test and UCSF set performance diminished in the other conditions (50% missing: 0.479 vs. 0.524; 100% missing: 0.2735 vs. 0.238). The WT subregion

Table 5: Comparison of the UCSF dataset and BraTS test set segmentation performance per tumor subregion.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
<i>BraTS training set</i>								
cross-validated	0.8599	0.8261	0.8685	0.8852	0.8057	0.7709	0.8109	0.8354
<i>BraTS test set: Real T1c</i>								
0% missing	0.7357	0.6754	0.7490	0.7826	0.6652	0.6110	0.6700	0.7145
50% missing	0.4794	0.3471	0.6719	0.4192	0.4138	0.3037	0.5752	0.3625
100% missing	0.2729	0.0760	0.6075	0.1353	0.2106	0.0515	0.4914	0.0889
<i>UCSF test set: Real T1c</i>								
0% missing	0.8396	0.8403	0.8333	0.8450	0.7828	0.7902	0.7687	0.7894
50% missing	0.5236	0.4191	0.7415	0.4102	0.4749	0.3933	0.6562	0.3753
100% missing	0.2379	0.0114	0.6454	0.0567	0.1936	0.0075	0.5382	0.0351
<i>Note:</i> The nnU-Net model was trained on exclusively real, complete data. Performance of the test datasets are evaluated under complete T1c, partial missing, and complete missing T1c conditions. Classes 1, 2, 3 correspond to tumor core (TC), whole tumor (WT), and enhancing tumor (ET) respectively.								

remained the most resistant to performance degradation, even with 100% missing T1c (BraTS: 0.608; UCSF: 0.645). Other subregions were severely affected by T1c missingness.

Tumor segmentation performance in the BraTS test set varied with respect to tumor size. Large tumors had the highest mean Dice scores (CV = 0.9089, test = 0.8167), followed by medium (CV = 0.8773, test = 0.7956), and small tumors (CV = 0.7580, test = 0.6646). The mean performance drop for medium to small tumors was substantially larger than from large to medium ($\Delta = 0.131$ vs. 0.021). WT was the most stable subregion. While Dice scores for TC in the test dropped by 0.5994 and ET by 0.6473 going from 0% to 100% missing, WT decreased minimally (0.7490 $\rightarrow$ 0.6075). This was observed across all conditions and sizes (see Tables 7, 8, and 9). No tumor size analysis was conducted for UCSF due to time constraints.

5.2.2 Synthetic Data Performance: RQ2

StyleGAN2-ADA was not evaluated for segmentation, as unconditional T1c synthetic scans cannot substitute for missing T1c. For SynDiff, only the final 100-slice configuration was used in all segmentation evaluations, as it

best replicated original conditions (Özbey et al., 2023), and achieved the best fidelity performance. Synthetic segmentation performance is available in Table 6.

Table 6: Segmentation performance per model on the BraTS 2023 dataset across tumor subregions.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
<i>BraTS training set</i>								
5-Fold cross-validation	0.8599	0.8261	0.8685	0.8852	0.8057	0.7709	0.8109	0.8354
<i>BraTS test set: Real T1c</i>								
0% missing	0.7357	0.6754	0.7490	0.7826	0.6652	0.6110	0.6700	0.7145
50% missing	0.4794	0.3471	0.6719	0.4192	0.4138	0.3037	0.5752	0.3625
100% missing	0.2729	0.0760	0.6075	0.1353	0.2106	0.0515	0.4914	0.0889
<i>BraTS test set: 50% Synthetic, 50% Real T1c</i>								
Pix2pixRAD	0.7011	0.6605	0.7351	0.7077	0.6256	0.5910	0.6524	0.6334
SynDiff	0.6770	0.6606	0.7151	0.6554	0.6004	0.5934	0.6288	0.5791
Custom Diffusion	0.5424	0.4161	0.6648	0.5463	0.4652	0.3578	0.5755	0.4624
<i>BraTS test set: 100% Synthetic T1c</i>								
Pix2pixRAD	0.6789	0.6609	0.7246	0.6513	0.5996	0.5889	0.6384	0.5715
SynDiff	0.6294	0.6463	0.6931	0.5489	0.5465	0.5775	0.5984	0.4638
Custom Diffusion	0.3784	0.1929	0.5965	0.3457	0.2954	0.1413	0.4940	0.2508
<i>Note: Classes 1, 2, 3 correspond to the mean segmentation performance on the tumor core (TC), whole tumor (WT), and enhancing tumor (ET) regions respectively. Mean refers to the averaged performance metric across all three tumor regions.</i>								

In both pix2pixRAD and SynDiff, the 50% synthetic condition achieved much higher Dice scores (pix2pixRAD: 0.7011; SynDiff: 0.6770) than the 50% missing (0.4794) and 100% missing (0.2729) T1c conditions. 100% synthetic T1c datasets for both models also outperformed both missing T1c conditions (pix2pixRAD: 0.6789; SynDiff: 0.6294). Pix2pixRAD outperformed SynDiff in both the 50% (0.7011 vs. 0.6770) and 100% (0.6789 vs. 0.6294) synthetic conditions. While custom diffusion performed poorly on all regions, it nonetheless outperformed all missing conditions in the 50% synthetic condition.

5.2.3 Tumor Sizes and Subregions: SQ3

All tumor size comparisons refer to Tables 7, 8, and 9. In both synthetic conditions, pix2pixRAD and SynDiff performed least well on ET relative to the 0% missing T1c condition (ET = 0.7826). Performance decreased significantly in ET from 50% to 100% synthetic (pix2pixRAD: 0.7077 $\rightarrow$ 0.6513; SynDiff: 0.6554 $\rightarrow$ 0.5489) compared to much smaller drops in TC and WT.

Table 7: Segmentation performance for large tumors across model variants.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
<i>BraTS training set</i>								
cross-validated	0.9089	0.8938	0.9148	0.9208	0.8649	0.8466	0.8703	0.8778
<i>BraTS test set: Real T1c</i>								
0% missing	0.8167	0.7513	0.8359	0.8629	0.7506	0.6911	0.7627	0.7980
50% missing	0.5244	0.3796	0.7611	0.4324	0.4570	0.3345	0.6625	0.3741
100% missing	0.3203	0.1078	0.7043	0.1487	0.2510	0.0751	0.5825	0.0955
<i>BraTS test set: 50% Synthetic, 50% Real T1c</i>								
Pix2pixRAD	0.7716	0.7248	0.8262	0.7639	0.6993	0.6598	0.7486	0.6895
SynDiff	0.7728	0.7471	0.8083	0.7631	0.6968	0.6847	0.7278	0.6780
Custom Diffusion	0.6112	0.4461	0.7663	0.6213	0.5288	0.3866	0.6769	0.5230
<i>BraTS test set: 100% Synthetic T1c</i>								
Pix2pixRAD	0.7523	0.7326	0.8172	0.7070	0.6759	0.6654	0.7360	0.6263
SynDiff	0.7498	0.7594	0.7908	0.6993	0.6667	0.6950	0.7032	0.6020
Custom Diffusion	0.4645	0.2303	0.6996	0.4636	0.3701	0.1729	0.5948	0.3427

Note: Classes 1, 2, 3 correspond to tumor core (TC), whole tumor (WT), and enhancing tumor (ET) respectively. Cross-validated BraTS training data are compared to segmentation of BraTS test data under real, 50% synthetic, and 100% synthetic T1c conditions. The total segmentation mean and subregion means for large tumors are shown.

Performance decreased going from large to small tumors, with ET under-performance holding across all tumor sizes. Across all conditions and sizes, WT was the most stable region.

Table 8: Segmentation performance for medium tumors across model variants.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
<i>BraTS training set</i>								
cross-validated	0.8773	0.8463	0.8824	0.9033	0.8228	0.7882	0.8261	0.8541
<i>BraTS test set: Real T1c</i>								
0% missing	0.7956	0.7526	0.8053	0.8289	0.7232	0.6866	0.7256	0.7573
50% missing	0.5607	0.4231	0.7373	0.5219	0.4913	0.3766	0.6414	0.4560
100% missing	0.3006	0.0684	0.6424	0.1911	0.2289	0.0446	0.5165	0.1255
<i>BraTS test set: 50% Synthetic, 50% Real T1c</i>								
Pix2pixRAD	0.7673	0.7403	0.7898	0.7718	0.6891	0.6681	0.7064	0.6927
SynDiff	0.7649	0.7432	0.7820	0.7695	0.6835	0.6695	0.6959	0.6851
Custom Diffusion	0.6207	0.4994	0.7296	0.6330	0.5393	0.4348	0.6415	0.5416
<i>BraTS test set: 100% Synthetic</i>								
Pix2pixRAD	0.7376	0.7258	0.7795	0.7076	0.6551	0.6527	0.6906	0.6219
SynDiff	0.7219	0.7428	0.7491	0.6739	0.6280	0.6639	0.6515	0.5686
Custom Diffusion	0.4280	0.2291	0.6388	0.4161	0.3326	0.1671	0.5302	0.3004

Note: Classes 1, 2, 3 correspond to tumor core (TC), whole tumor (WT), and enhancing tumor (ET) respectively. Cross-validated BraTS training data are compared to segmentation of BraTS test data under real, 50% synthetic, and 100% synthetic T1c conditions. The total segmentation mean and subregion means for medium tumors are shown.

While mean Dice scores for TC in the test dropped by 0.5994 and ET by 0.6473, WT decreased minimally (0.7490 $\rightarrow$ 0.6075). Additionally, 100% synthetic pix2pixRAD had a similar Dice score to the 0% missing condition for WT (0.7246 vs. 0.7490). 100% synthetic custom diffusion achieved higher Dice for ET than the 100% missing baseline (0.3457 vs. 0.1353).

Table 9: Segmentation performance for small tumors across model variants.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
<i>BraTS training set</i>								
cross-validated	0.7580	0.6724	0.7941	0.8076	0.6890	0.6064	0.7153	0.7454
<i>BraTS test set: Real T1c</i>								
0% missing	0.6646	0.6189	0.6517	0.7233	0.5821	0.5452	0.5560	0.6450
50% missing	0.4774	0.3514	0.5758	0.5050	0.4016	0.3018	0.4680	0.4350
100% missing	0.2535	0.0615	0.5265	0.1726	0.1904	0.0399	0.4113	0.1201
<i>BraTS test set: 50% Synthetic, 50% Real T1c</i>								
Pix2pixRAD	0.6200	0.5975	0.6288	0.6336	0.5386	0.5236	0.5313	0.5609
SynDiff	0.5989	0.6231	0.5935	0.5801	0.5132	0.5478	0.4890	0.5027
Custom Diffusion	0.4875	0.4051	0.5385	0.5190	0.4084	0.3436	0.4379	0.4436
<i>BraTS test set: 100% Synthetic T1c</i>								
Pix2pixRAD	0.5742	0.5720	0.6140	0.5366	0.4911	0.4937	0.5145	0.4650
SynDiff	0.5308	0.5799	0.5644	0.4480	0.4373	0.4998	0.4530	0.3591
Custom Diffusion	0.2836	0.1262	0.4717	0.2528	0.2114	0.0884	0.3685	0.1772
<i>Note:</i> Classes 1, 2, 3 correspond to tumor core (TC), whole tumor (WT), and enhancing tumor (ET) respectively. Cross-validated BraTS training data are compared to segmentation of BraTS test data under real, 50% synthetic, and 100% synthetic T1c conditions. The total segmentation mean and subregion means for small tumors are shown.								

5.3 Segmentation Generalization: SQ4

Pix2pixRAD and SynDiff outperformed missing T1c conditions with UCSF (pix2pixRAD: 0.6480; SynDiff: 0.6824). SynDiff outperformed pix2pixRAD across all tumor regions on UCSF (Table 9). In the ET region, pix2pixRAD suffered a 22% drop in Dice score between the BraTS and UCSF dataset (0.7077 $\rightarrow$ 0.5494). WT remained resistant to decline.

100% synthetic SynDiff outperformed 50% synthetic SynDiff (0.7023 vs. 0.6824) and both pix2pixRAD conditions (50%: 0.6480; 100%: 0.4482). From 50% to 100% synthetic SynDiff, TC segmentation improved (0.6614 $\rightarrow$ 0.7326), with WT and ET remaining stable. Dice scores decreased across all regions from 50% to 100% synthetic Pix2pixRAD (mean: 0.6480 $\rightarrow$ 0.4482).

Custom diffusion again performed the most poorly in both conditions, though still performing better than 100% missing T1c.

Table 10: Segmentation performance on the UCSF dataset per model across tumor subregions.

Model	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
<i>BraTS training set</i>								
cross-validation	0.8599	0.8261	0.8685	0.8852	0.8057	0.7709	0.8109	0.8354
<i>UCSF dataset: Real T1c</i>								
0% missing	0.8396	0.8403	0.8333	0.8450	0.7828	0.7902	0.7687	0.7894
50% missing	0.5236	0.4191	0.7415	0.4102	0.4749	0.3933	0.6562	0.3753
100% missing	0.2379	0.0114	0.6454	0.0567	0.1936	0.0075	0.5382	0.0351
<i>UCSF dataset: 50% Synthetic, 50% real T1c</i>								
Pix2pixRAD	0.6480	0.6229	0.7715	0.5494	0.5866	0.5737	0.6948	0.4914
SynDiff	0.6824	0.6614	0.7765	0.6094	0.6240	0.6178	0.7004	0.5539
Custom Diffusion	0.5779	0.5348	0.6927	0.5061	0.5133	0.4823	0.6157	0.4420
<i>UCSF dataset: 100% Synthetic T1c</i>								
Pix2pixRAD	0.4482	0.3974	0.7044	0.2428	0.3796	0.3464	0.6133	0.1792
SynDiff	0.7023	0.7326	0.7722	0.6022	0.6280	0.6759	0.6896	0.5185
Custom Diffusion	0.3611	0.2426	0.5617	0.2790	0.2884	0.1874	0.4726	0.2053

Note: This table shows results from the segmentation generalization experiment, with UCSF used in place of BraTS. Synthetic data were used to supplement T1c from the UCSF dataset in the model conditions. Classes 1, 2, 3 correspond to the mean segmentation performance on the tumor core (TC), whole tumor (WT), and enhancing tumor (ET) regions respectively. Mean refers to the averaged performance metric across all three tumor regions.

5.4 Error Analysis

Segmentation discrepancy between the BraTS test set and the other datasets was investigated. Full results are reported in Appendix 6.14 due to formatting constraints, with key findings summarized below.

FN, FP and complete miss rates (Dice = 0) were substantially higher in the BraTS test set than in BraTS train or UCSF (Appendix 6.14, Table 14). This was most pronounced in small tumors: NCR (FN: 0.514; FP: 0.351; complete miss: 18.8%), edema (FN: 0.315; FP: 0.332; complete miss: 0%), and ET (FN: 0.384; FP: 0.220; complete miss: 5.3%). Figure 11 depicts

complete miss rates. Segmentation failure ($\text{Dice} < 0.3$) was associated with significantly elevated CNR and SNR in T2w and FLAIR scans across all datasets (Appendix 6.14, Table 15).

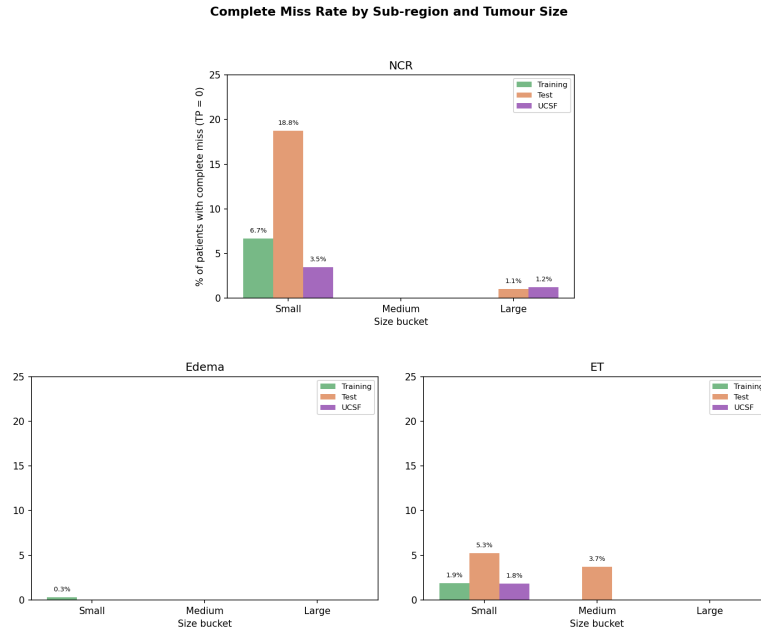


Figure 11: Complete miss rate ($\text{TP} = 0$) between all datasets, tumor subregions, and tumor sizes.

The BraTS test set contained more large tumors than the BraTS train and UCSF datasets – e.g., 79.1% large NCR in BraTS test, 20.7% in UCSF, and 44.6% in the BraTS train set (Figure 12). UCSF had more small tumors than other datasets – e.g., 52.9% small edema tumors compared to BraTS test (2.8%) and train (33%) datasets.

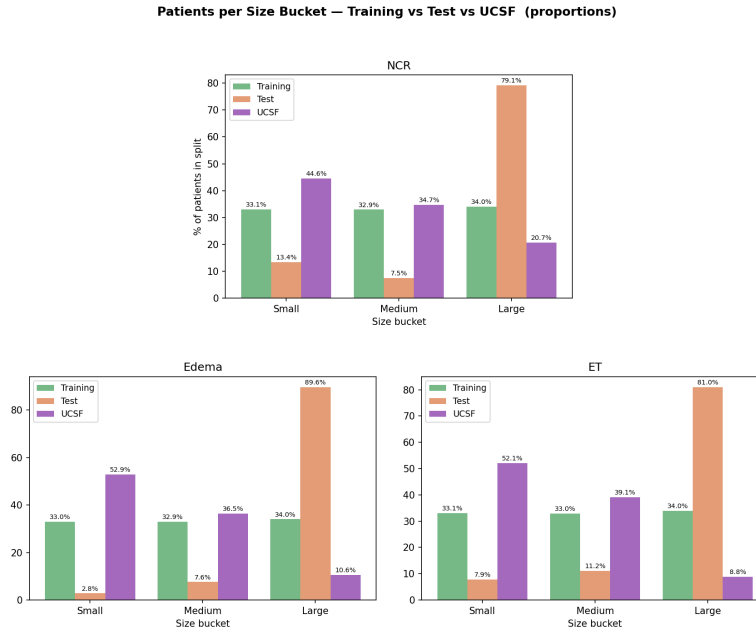


Figure 12: The proportion of large, medium, and small tumors per subregion, per dataset. Tumor size was determined through binning by quantiles, with reference to tumors present in the BraTS training data.

6 DISCUSSION

This study investigated to what degree state-of-the-art generative models could synthesize high-fidelity 2D T1c scans that could substitute for missing T1c in downstream glioma segmentation. Fidelity and segmentation performance analysis revealed T1c synthesis is achievable with substantial variation based on conditioning strategy, model type, tumor size, subregion and dataset. Overall, synthetic T1c substitution improved segmentation over missing T1c conditions, though not achieving performance of a complete dataset, a finding corroborated in prior work (Raut et al., 2024; Usman Akbar et al., 2024).

6.1 RQ1: To what extent can state-of-the-art generative models generate T1c images that are comparable in quality to ground truth T1c?

Both pix2pixRAD and SynDiff achieved competitive fidelity across distributional, pixel-wise and perceptual dimensions, suggesting high-fidelity 2D T1c can be generated with appropriate conditioning. For example, structural/pixel-wise fidelity scores were comparable to those reported in

the original studies for T1w synthesis: SynDiff achieved SSIM: 0.9415 on T1c, compared to SSIM: 0.9379 on T1w (Özbey et al., 2023); and pix2pixRAD achieved SSIM: 0.9509, comparable to SSIM: 0.92 reported for T1w (Conte et al., 2021).

However, initial model training revealed the difficulty of T1c synthesis, with frequent mode collapse and tumor learning failure when conditioning on non-contrast inputs alone (Appendices 6.11, 6.12). A multi-input conditioning approach with tumor spatial priors was ultimately successful for pix2pixRAD, consistent with research finding multi-input approaches to better handle contrast-dependent modalities (Dong et al., 2026; Kim & Park, 2024). Embedding tumor spatial priors into SynDiff’s single input channel improved learning, analogous to studies using segmentation masks as conditioning inputs (Bhattacharya et al., 2025; Mirowski & Fabijańska, 2026). However, custom diffusion’s failure to synthesize T1c with segmentation masks alone suggests spatial priors are insufficient for 2D T1c conditioned synthesis, which is challenging enough using non-contrast scans (Dong et al., 2026; Eidex et al., 2025; Tabassum et al., 2024).

6.2 SQ1: *To what extent can an unconditional model learn the T1c distribution?*

StyleGAN2-ADA’s strong distributional fidelity compared to conditional models indicates that the T1c distribution is learnable to a meaningful extent independent of conditioning. This suggests conditional synthesis failures reflect insufficient guidance rather than a fundamental unlearnability of T1c. Additionally, StyleGAN2-ADA’s distribution metrics compared favorably to its scores reported in non-contrast synthesis: FID = 20.44 and KID = 0.006 (present study), compared to FID = 26.92 and KID = 0.026 (Lai et al., 2025); and FID = 58.11 with KID = 0.009 (Tariq et al., 2023).

6.3 SQ2: *To what extent can models generalize to an external dataset?*

SynDiff demonstrated better distributional and perceptual fidelity in BraTS test, consistent with evidence that diffusion models produce more diverse image distributions than GANs (Müller-Franzes et al., 2023). Pix2pixRAD’s stronger pixel-wise learning was consistent with GANs rigid structural learning (Isola et al., 2017), explaining the better generalizability of SynDiff to UCSF than pix2pixRAD: pix2pixRAD’s rigid learning may have led to its decrease in distributional fidelity, and SynDiff’s stable fidelity suggests superior distribution learning allowed greater generalizability.

StyleGAN2-ADA’s distributional fidelity decreased substantially on the UCSF dataset, which may reflect distributional dissimilarity between datasets rather than deficient unconditional learning. BraTS was collected

across 19 different institutions using diverse scanners and protocols (Bonato et al., 2025), while UCSF employed a single scanner (Calabrese et al., 2023), likely producing a narrower image distribution. Since StyleGAN2-ADA cannot compensate for this mismatch through conditioning, this suggests unconditional T1c synthesis is sensitive to scanner/protocol diversity. A model trained on multi-institutional data may learn a broader distribution, which may not align well to a single-institution context. This highlights scanner/protocol heterogeneity as a potentially meaningful variable in unconditional T1c synthesis.

6.4 RQ2: *To what degree can synthetic conditional T1c images substitute for missing T1c in downstream segmentation?*

Synthetic T1c substitution with pix2pixRAD and SynDiff outperformed missing T1c conditions, demonstrating that state-of-the-art diffusion and GAN models can be adapted to generate potentially clinically useful T1c scans. Performance remained below the 0% missing T1c condition, consistent with prior synthetic T1c use in segmentation (Raut et al., 2024; Usman Akbar et al., 2024). Pix2pixRAD outperformed SynDiff despite inferior distributional and perceptual fidelity, suggesting tumor region accuracy may matter more than holistic fidelity for downstream segmentation (Raut et al., 2024). This aligns with evidence that shows image quality metrics may be insensitive to local anatomical details critical for accurate segmentation (Deo et al., 2025). Therefore, pix2pixRAD’s rigid pixel-wise mapping may have produced tumor regions that initially better corresponded to nnU-Net expectations.

6.5 SQ3: *To what extent does synthetic T1c segmentation performance vary by tumor size/region?*

Segmentation performance decreased as tumor regions became smaller, consistent with previous segmentation research (Erdur et al., 2023; Raut et al., 2024). Specifically, the undersegmentation of ET for all models revealed insufficient tumor enhancement pattern learning as a key weakness in T1c synthesis approaches, even with tumor spatial prior conditioning. This may be due to the high degree of heterogeneity of gadolinium enhancement (Osman & Tamam, 2023; Warntjes et al., 2018), representing potential disparate impact for segmentation supplementation in glioma patients – especially those with pronounced ET (e.g., HGG).

However, despite this decreased performance, both models improved on the Raut et al. (2024) baseline, which found the best performing T1c substitution condition achieved Dice = 0.12 ($\pm$ 0.11) on the ET region

(100% synthetic pix2pixRAD = 0.65; SynDiff = 0.55). Additionally, both pix2pixRAD and SynDiff outperformed missing T1c conditions in TC. Because TC incorporates NCR and ET, this suggests the models could synthesize robust enough NCR regions to compensate for poor ET estimation. Therefore, despite ET learning challenges, synthetic T1c may offer genuine clinical utility for non-enhancing tumor tissues. Incidentally, WT stability across models and conditions reflects known dependence on T2w/FLAIR edema visualization rather than T1c (Stall et al., 2010).

6.6 SQ4: To what extent can synthetic T1c segmentation performance generalize to an external dataset?

Pix2pixRAD and SynDiff outperformed missing T1c conditions on UCSF, but robustness was contingent on distributional rather than pixel-wise learning. Pix2pixRAD’s performance decreased across all tumor regions, consistent with rigid mappings in GANs (Isola et al., 2017). This was most pronounced in ET, demonstrating pixel-wise learning could not fully capture heterogeneous enhancement patterns that vary widely between patients (Osman & Tamam, 2023; Warntjes et al., 2018).

Counterintuitively, 100% synthetic SynDiff outperformed 50% synthetic conditions on UCSF. This finding may reflect distributional dissimilarity between datasets, as previous research notes that performance can be influenced both by distributional consistency and image fidelity (Jahagirdar & Joshi, 2025; Usman Akbar et al., 2024). Since UCSF data came from a single institutional scanner, BraTS-trained synthetic T1c was likely more distributionally similar to BraTS than UCSF – supported by StyleGAN2-ADA’s decreased distributional fidelity metrics on UCSF. The segmentation model may have encountered a distributional mismatch in the 50% synthetic condition not present in the 100% synthetic condition, leading to reduced segmentation.

6.7 Error Analysis

The train-test performance gap appeared to be driven by chance overrepresentation of atypical glioma phenotypes in the test set, rather than tumor size class imbalance. Failed cases showed a higher concentration of elevated T2w and FLAIR signal metrics, which were overrepresented in the test set. This may indicate more extensive, diffuse edema obscuring other subregion boundaries (Claes et al., 2007; Schmitz-Abecassis et al., 2025; Sengupta et al., 2018). High FP rates in catastrophic failures (Dice < 0.3) confirmed other tumor regions were mislabeled as edema. Since diffuse

glioma tumors may be challenging to delineate, a higher concentration of them may explain the performance discrepancy (Wan et al., 2024).

6.8 *Societal and Scientific Implications*

This study establishes that state-of-the-art generative models developed for non-contrast 2D brain MRI synthesis can be meaningfully adapted to T1c, and that synthetic T1c substitution offers genuine but partial clinical utility, with small tumors and enhancing tumor regions identified as primary synthesis challenges. Fidelity and segmentation results provide systematic baselines for 2D T1c synthesis across GANs and DDPMs, addressing underrepresentation of T1c in MRI synthesis (Dong et al., 2026; Eidex et al., 2025; M. S. K. Luu et al., 2025), incomplete fidelity evaluation (Deo et al., 2025; Dohmen et al., 2025), assessment of image fidelity without downstream clinical utility testing (Dayarathna et al., 2024; Wu et al., 2024), over-representation of GANs in downstream tasks compared to diffusion models (Azad et al., 2026; Dayarathna et al., 2024; Eidex et al., 2026); and limited generalizability assessment (Ibrahim et al., 2025; Suleman et al., 2025).

The societal implications of this study involve the improvement of T1c access for glioma patients unable to obtain contrast scans, whether due to resource constraints (Kim & Park, 2024) or contraindications like kidney disease complications and pregnancy risks (Bird et al., 2019; Schieda et al., 2018).

Considering ET delineation is critical for glioma assessment, especially in highly malignant HGGs with poor prognosis (Hirschler et al., 2023; Unkelbach et al., 2014), the persistent failure to accurately synthesize ET represents a meaningful limitation in clinical utility. The misapplication of such T1c synthesis models risks disparate negative impact on glioma patients who present with pronounced ET regions. Therefore, synthetic T1c use from these models may be better suited for non-enhancing tumor tissue segmentation, and clinical use should be accompanied by explicit communication to clinicians that ET assessments may be unreliable.

6.9 *Limitations and Future Directions*

This study had several limitations. The train-test split was unstratified, which may have contributed to an over-representation of atypical glioma phenotypes in the test set. All models were trained on 2D axial slices rather than 3D volumes, limiting spatial context and direct clinical applicability. Image fidelity evaluation would have been strengthened with blind rating by radiologists, and segmentation assessment would have benefited by the

inclusion of Hausdorff’s distance, which captures worst-case boundary errors that Dice/IoU cannot measure (Aydin et al., 2021).

All data were heavily preprocessed – including skull-stripping, co-registration to a standardized anatomical template, resampling to a uniform resolution, and intensity normalization (Bakas et al., 2017c; Calabrese et al., 2023). In practice, scans may not undergo extensive preprocessing, potentially limiting direct applicability to real-world clinical contexts. Furthermore, results suggested that different acquisition protocols may influence underlying image distributions, which may meaningfully affect synthesis performance. Therefore, the use and interpretation of models applied to scans from institutions with different scanning protocols should be approached with caution. Finally, both datasets were representative of wealthier, western demographics, limiting generalizability to under-resourced settings where T1c scarcity is more acute.

Future studies should stratify by tumor size and grade (if labeled) to avoid chance imbalances in glioma type representation, and extend T1c conditioning strategies to 3D architectures like LDM (Eidex et al., 2026; Pinaya et al., 2023; Rombach et al., 2022) or pix2pixNifTI (Raut et al., 2024) to improve 3D T1c synthesis. Additionally, future studies could examine generalization robustness of unconditional T1c synthesis with DDPMs, given the tendency of diffusion models toward more diverse image generation (Azad et al., 2026; Dorjsembe et al., 2024; Müller-Franzes et al., 2023).

Research should be extended to diverse patient populations such as BraTS-Africa (Adewole et al., 2025), or to pediatric gliomas. Fine-tuning nnU-Net with mixed synthetic T1c data may improve evaluation performance without compromising performance on real scans (Pani & Chawla, 2024; Usman Akbar et al., 2024). Finally, mask-free conditioning and improved enhancement learning are identified as key steps for more complete synthetic T1c clinical utility in glioma patients.

6.10 Conclusion

T1c scans are critical for glioma assessment, yet frequently absent from clinical and research datasets, adversely affecting patient care and segmentation model use. This study demonstrated state-of-the-art GAN and diffusion models can be adapted to generate high-fidelity 2D T1c scans that improved segmentation over missing T1c conditions. However, tumor information was necessary but not sufficient to guide high-fidelity synthesis, and enhancing tumor regions were consistently challenging to synthesize. With improvements to conditioning strategy and enhancement learning,

synthetic T_{1c} generation could foreseeably expand access to crucial glioma assessment for patients unable to obtain contrast scans.

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APPENDIX A

Table 11: Software dependencies across models

Dependency	Pix2PixRAD	StyleGAN2-ADA	SynDiff	Custom Diffusion	nnU-Net
Python	3.10.0	3.8.20	3.8.20	3.10.19	3.10.19
PyTorch	2.10.0	1.7.1	1.7.1	2.0.1	2.5.1
TorchVision	0.25.0	–	0.8.2	0.15.2	0.20.1
TorchAudio	–	–	–	–	2.5.1
TensorFlow	2.20.0	–	–	–	–
Keras	3.12.0	–	–	–	–
CUDA wheel	–	11.0	11.0	–	12.1
CUDA Runtime	12.8.90	–	–	11.7.99	12.1.105
CUDA Toolkit	–	–	11.0.3	–	–
cuDNN	9.10.2.21	–	8.0.5	8.5.0.96	9.1.0.70
NumPy	2.2.6	1.24.4	1.24.3	1.26.4	2.2.6
SciPy	1.15.3	1.10.1	1.10.1	1.15.2	1.15.3
Pandas	–	–	2.0.3	–	2.3.3
Pillow	12.1.0	10.4.0	9.3.0	–	12.0.0
Matplotlib	–	–	3.7.5	–	3.10.8
Seaborn	–	–	0.13.2	–	0.13.2
scikit-image	–	–	0.21.0	–	0.25.2
scikit-learn	–	–	–	–	1.7.2
imageio	–	–	2.35.1	–	–
imageio-ffmpeg	–	0.4.3	–	–	–
h5py	–	–	3.11.0	–	–
MONAI	–	–	–	1.4.0	–
MONAI-Generative	–	–	–	0.2.3	–
NiBabel	–	–	–	5.3.3	5.4.0
SimpleITK	–	–	–	–	2.5.3
nnunetv2	–	–	–	–	2.6.4
pypng	–	0.1.4	–	–	–
Ninja	–	1.13.0	1.13.1	–	–
click	–	8.1.8	–	–	–

Dependency	Pix2PixRAD	StyleGAN2-ADA	SynDiff	Custom Diffusion	nnU-Net
requests	–	2.32.4	–	–	–
tqdm	–	4.67.1	–	–	–

6.11 A1: Pix2pixRAD

ARCHITECTURE The current study employed the same baseline architecture as Conte et al. (2021). The Pix2pixRAD generator consists of 8 convolutional downsampling layers composing a single downstack, and 8 convolutional upsampling layers composing an upstack. The function of the downstack is to compress the input image into a compact feature representation to encourage the model to learn foundational structural features, whereas the upstack forces the model to reconstruct the image to its 256 x 256 resolution using the features learned from the downstack. The generator also makes use of batch normalization to stabilize training, skip layers to pass along feature information, and leakyReLU to avoid disappearing gradients by allowing small negative gradients. Figure 13 visualizes the basic generator architecture.

The discriminator architecture is composed of three successive downsampling layers, and likewise makes use of batch normalization and leakyReLU. In addition, zero padding is applied to ensure no edge information is lost between convolutions. A representation of the discriminator is shown in Figure 14. The model originally used binary cross entropy adversarial loss, training the discriminator to distinguish between real and synthetic images using the probability of synthetic vs. real. Finally, the model used an Adam generator and discriminator optimizer of 2e-4 and originally employed a λ L1 loss of 100, which functions to strongly enforce pixelwise similarity between the predicted and target image.

TRAINING Initial training was implemented for 300 epochs as specified by Conte et al. (2021) using the following hyperparameters: Binary cross entropy loss, λ -L1= 200, Adam optimizer with generator and discriminator loss = 2e-4, β_1 = 0.5, β_2 = 0.99, and minibatch stochastic gradient descent with batch size = 1. The initial model was trained in a translation task, from T1w $\rightarrow$ T1c, using input of T1w scan across 3 channels.

Model training was marked by frequent mode collapse, along with rapid discriminator loss decrease coupled with generator loss stagnation. The discriminator learning rate was incrementally decreased to 5e-5, mitigating mode collapse—however, outputs were consistently blurry and anatomically undifferentiated. To reduce over-reliance on penalizing pixel-

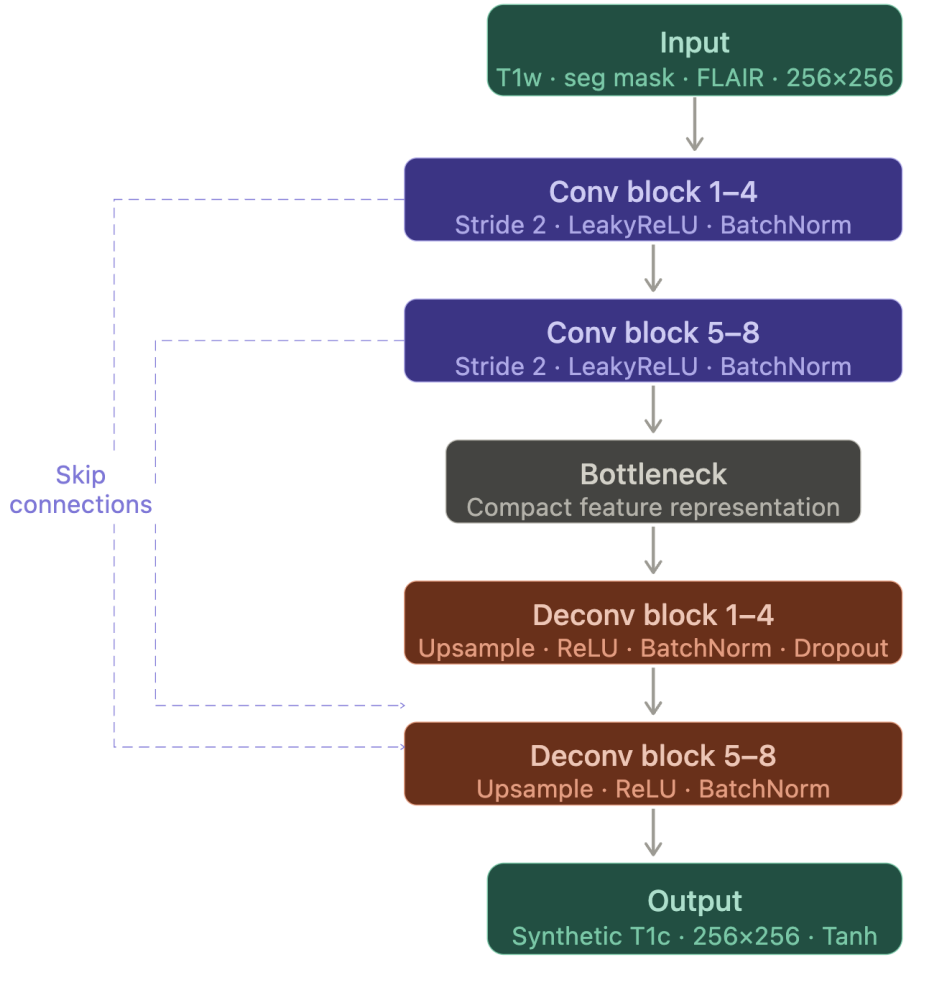


Figure 13: A representation of the pix2pix generator architecture.

wise differences and encourage the generator to produce a greater degree of contrast between brain regions, λ -L1 was decreased incrementally to 3. Though helpful, these changes did not help the model to sufficiently learn mappings to tumor regions.

Reasoning that T1w alone did not provide sufficient tumor-related information, the model was trained with a modified input, containing T1w in channels 1 and 3, and a segmentation mask scaled to $[0, 255]$ in channel 2. Despite marginal improvements in image quality, outputs still lacked anatomical contrast. Therefore, perceptual loss (VGG-16) and gradient loss were introduced to encourage image sharpness and edge-level differentiation, since decreasing λ -L1 alone was insufficient (Johnson et al., 2016; Ma et al., 2020).

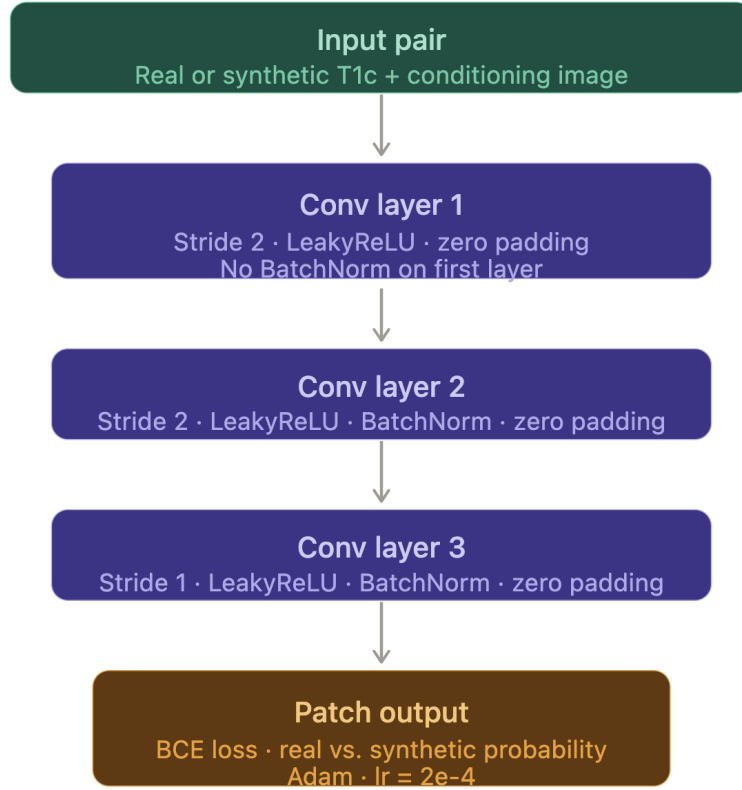


Figure 14: A representation of the pix2pix discriminator architecture.

To further stabilize training, pretrained weights were used from the Cityscape dataset (Cordts et al., 2016). In spite of the domain gap, this stabilized training dynamics significantly, though the model still suffered from blurred brain anatomy. Finally, the input was further modified such that the channels contained T1w, the scaled mask, and the corresponding FLAIR scan, respectively. This was done to encourage learning tumor-relevant information and anatomical detail that is not fully present in both T1w scans and segmentation masks. Additionally, binary cross-entropy loss was replaced with least-squares loss (LSGAN) (X. Mao et al., 2017). This was done to avoid saturated gradients common with binary cross-entropy loss, which resulted in discriminator loss consistently outpacing generator loss.

The final model consisted of pretrained cityscape weights, an input of T1w, scaled segmentation masks, and FLAIR. Furthermore, the following hyperparameters were implemented: least-squares loss, λ -L1 = 1, batch size = 2, generator loss = 2e-4, discriminator loss = 5e-5, β L1 = 0.5, λ -

perceptual = 10, λ gradient = 10, epochs = 85. A comprehensive overview of training is shown in Table 12.

Table 12: Pix2PixRAD hyperparameter configurations across training stages

Parameter	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Exp. 5	Exp. 6	Exp. 7
Input Channels	T1w x 3	T1w x 3	T1w, seg, T1w	T1w, seg, T1w	T1w, seg, FLAIR	T1w, seg, FLAIR	T1w, seg, FLAIR
Epochs	300	–	–	–	–	–	85
Loss Function (BCE)	✓	✓	✓	✓	✓	✓	LSGAN
λ L1	100	25	3	3	3	1	1
λ Perceptual	–	–	–	–	10	10	10
λ Gradient	–	–	–	–	10	10	10
Generator LR	2e-4	2e-4	2e-4	1e-4	1e-4	1e-4	1e-4
Discriminator LR	2e-4	5e-5	5e-5	5e-5	5e-5	5e-5	5e-5
Batch Size	1	1	1	1	2	2	2
Pretrained Weights	–	–	–	Cityscape	Cityscape	Cityscape	Cityscape
Assessment	Mode collapse: black outputs; near zero discriminator loss	Mode collapse; grainy images with minimal anatomical differentiation	Reduced blurring; improved but poor anatomical learning.	Image quality improvement; still lacking sufficient anatomical contrast	Significant improvement of anatomy learning; images remain blurry	Stabilized training dynamics; improved image sharpness	Stable training; successful learning of tumor anatomy

Note: LR = Learning Rate; BCE = Binary Cross-Entropy; LSGAN = Least Squares GAN; Loss function was MSE unless otherwise specified; λ Perceptual = VGG-16; seg = modified segmentation mask. Epochs less than 50 were not specified, given early model failure. More configurations were tried besides those listed in the table, but adjustments were deemed minor enough to not warrant inclusion.

6.12 A2: SynDiff

ARCHITECTURE The current study employed the same baseline architecture as Özbey et al. (2023). SynDiff employs a hybrid architecture, training both a non-diffusive and diffusive module jointly. The purpose of the non-diffusive module is to estimate an input image paired to each target image, in order to better guide the diffusion model. It consists of two pairs of GANs, the generators of which are built on a ResNet backbone, with 3 encoding blocks, 6 residual blocks, and 3 decoding blocks - an example of which can be seen in Figure 15. Both GANs are trained with adversarial loss, while also applying cycle-consistency loss to ensure that the model can learn to translate in both directions – i.e., A B A (Figure 16).

The diffusive module consists of two generators with an underlying UNet architecture consisting of 6 encoding and decoding blocks, as well as residual subblocks and temporal/latent embeddings to adjust the degree of applied noise based on the current position along the diffusion process. These are paired with discriminators which are comprised of 6 blocks of 2 convolutional layers with downsampling. The diffusion process comprises of 4 reverse diffusion steps, with 1000 steps and a step size of $k = 250$. An example of the diffusive module can be seen in Figure 17.

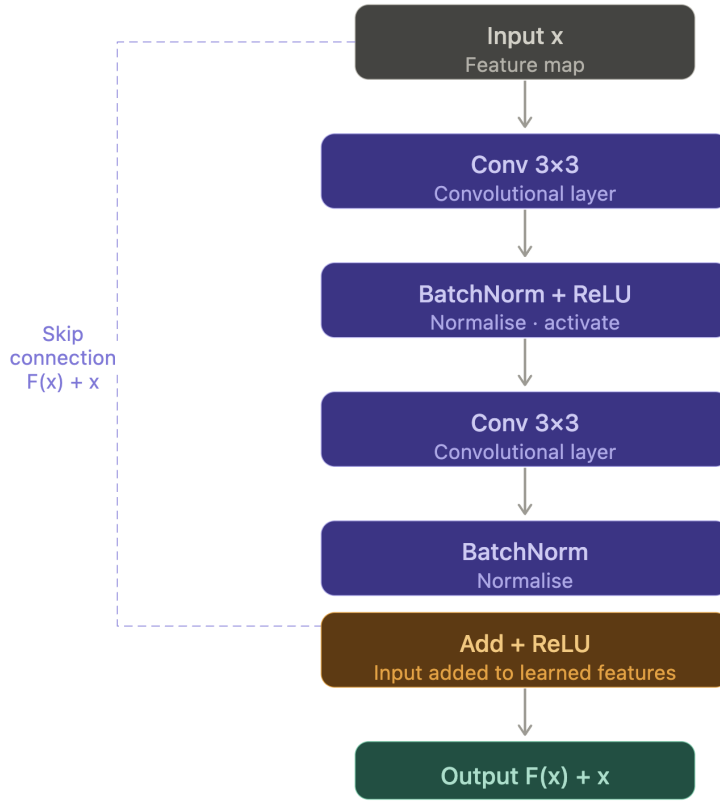


Figure 15: A representation of a ResNet architecture.

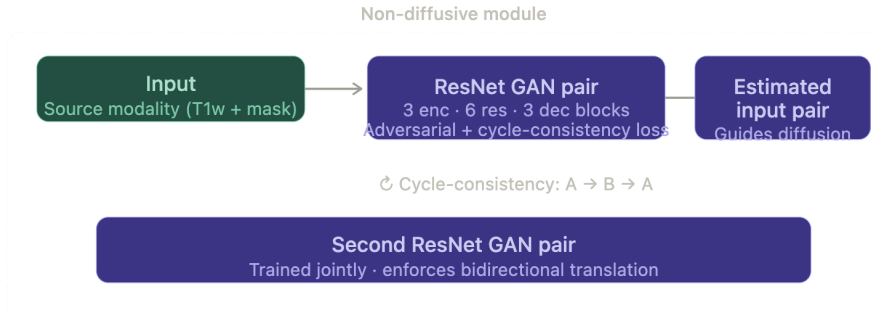


Figure 16: A representation of the SynDiff non-diffusive module architecture.

As the task of this study was to synthesize T1c scans in one direction, only the generator performing this unidirectional mapping was employed. By extension, only the relevant diffusion generator was used at inference.

TRAINING This study made use of the 50-epoch pretrained weights for the T2w → T1w translation task available on the SynDiff GitHub repo (Özbey et al., 2023). This was done both to save computing resources,

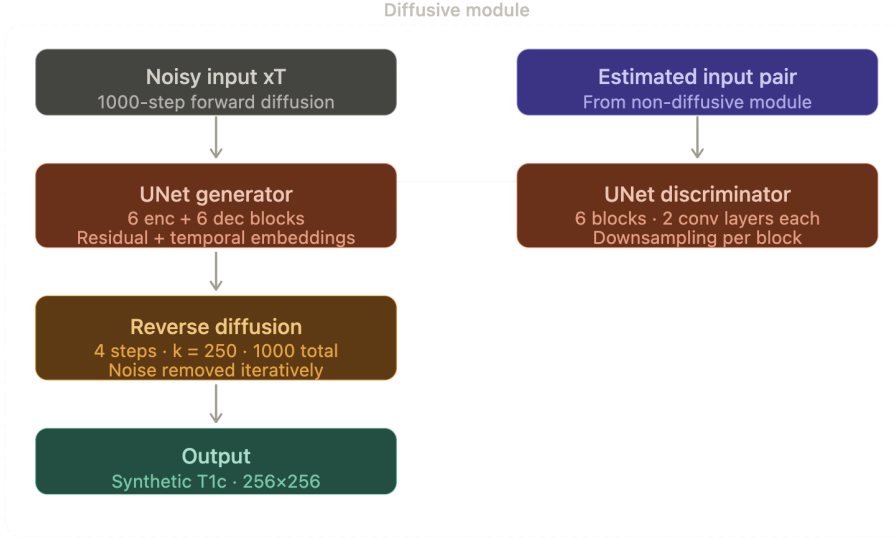


Figure 17: A representation of the SynDiff diffusive module architecture.

and to take advantage of learned brain MRI mappings. Additionally, this specific direction was chosen due to the relatedness of T1w and T1c scans, and it was reasoned that the pretrained model had learned to generate anatomical features consistent with the intensity patterns present in T1 scans. Batch size was initially increased from 1 to 2 to speed up and stabilize training.

In order to determine which scan type would provide the most useful anatomical information for a translation task to T1c, this study tested four different inputs: FLAIR, T1w, and two versions of a modified T1w scan. The modified scans consisted of T1w scans concatenated with a scaled segmentation mask ($[0,3] \rightarrow [0,255]$) such that pixel intensities were transformed according to the following equation: modified T1w = T1w + α * scaled segmentation mask, with intensities of $\alpha = 0.1$, and $\alpha = 0.3$.

FLAIR was chosen due to its similarity to the original input of T2w scans, as well as the fact that it contained tumor-related information and rich anatomical detail. Though pixel intensity distributions differ widely between FLAIR and T1c, it was reasoned that they differ in similar enough ways from T2w and T1w to warrant testing. T1w was selected considering it is in a similar domain to T1c, so the model may have an easier time with similar input and target pixel intensity distributions. Finally, T1w scans modified with scaled segmentation masks were used as an exploratory test to see if the model would benefit from additional tumor related information that would otherwise not be present in T1w scans alone.

All models were initially trained for 5 epochs with early stopping of patience 3 based on highest PSNR score. The best training checkpoints from each model were selected, and underwent further training with generator learning rate = $8e-5$ and discriminator learning rate = $5e-5$ to encourage the learning of subtle anatomical details. Each model was trained for a further 15 epochs with patience of 5. All initial model versions were trained and tested on the 15-slice dataset, while the best performing model was trained and tested on the 100-slice per patient dataset. Fidelity scores of varying configurations are shown in Table 2.

6.13 A3: Custom Diffusion

The custom diffusion model presented by Mirowski and Fabijańska (2026) is constructed from a U-Net architecture which progresses through alternating downsampling and attention blocks (encoder), followed by alternating upsampling and attention blocks (decoder). As a UNet, the model employs skip connections between sequential encoding and decoding blocks, allowing it to pass on detailed feature information that may otherwise be missed through downsampling. The addition of attention layers helps to preserve global context, allowing the model to retain anatomical information through successive down/upsampling. An example U-Net is shown in Figure 18.

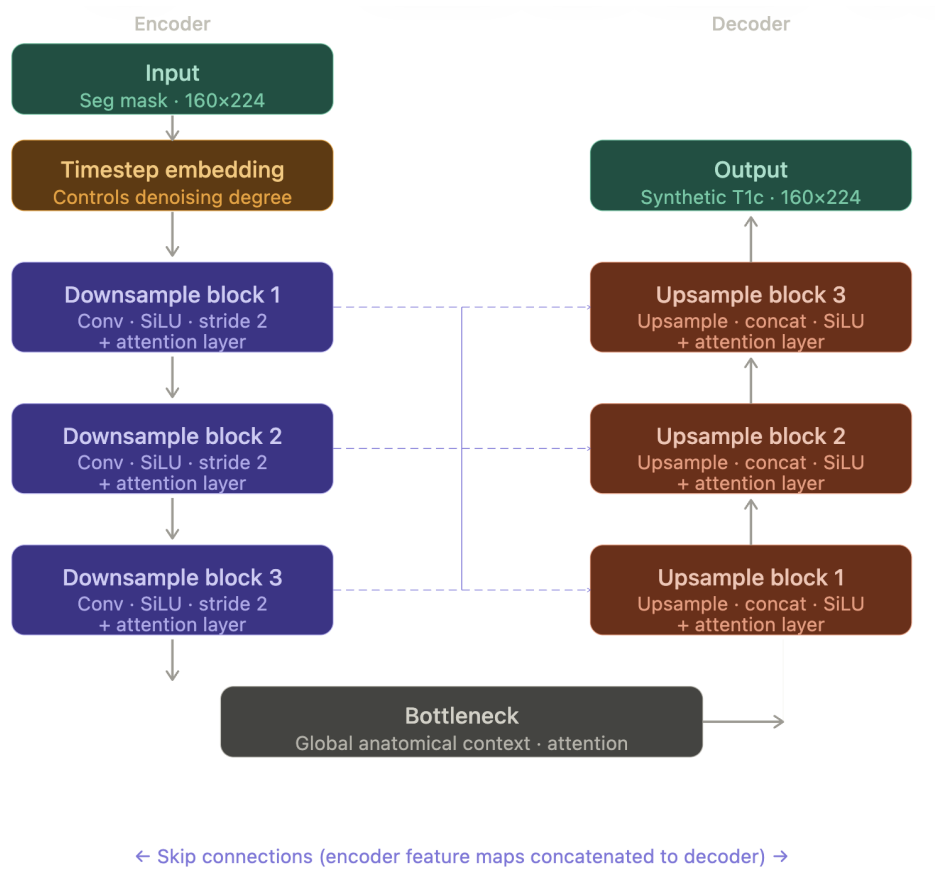


Figure 18: A representation of the U-Net employed in the custom diffusion architecture.

As with Syndiff, custom diffusion utilizes timestep conditioning so that the model can apply the correct degree of denoising depending on timestep. Interestingly, in place of ReLU, custom diffusion uses sigmoid linear (SiLU) activation, which allows for a slight negative gradient and a smoother gradient profile without risking vanishing gradients. Finally, a dropout rate of 0.3 was used to prevent overfitting—especially important given the limited amount of medical training data.

Additional model information can be viewed in Table 13, together with intermediate training output samples in Figure 19.

Table 13: Custom diffusion hyperparameter configurations across training stages.

Parameter	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Exp. 5	Exp. 6	Exp. 7	Exp. 8
Corresponding Image	–	a	b	c	d	f	g	h
Conditioned Input	seg	seg	FLAIR	seg	seg	seg	seg	seg
T1c Normalization	[0,1]	[0,1]	[0,1]	[0,1]	[0,1]	[-1,1]	[-1,1]	[-1,1]
LR	1e-4	1e-4	1e-4	1e-4	1e-5	1e-5	1e-5	1e-4
Epochs	200	–	–	–	100	175	200	200
LR Schedule (Linear)	✓	Cosine	✓	✓	✓	✓	✓	✓
Noise Loss (MSE)	✓	✓	✓	SSIM	✓	✓	✓	✓
Intensity Loss	–	–	–	–	0.02	0.02	–	–
Clamping Loss	–	–	–	–	0.04	0.04	–	–
Clamping Range	–	–	–	–	–	[-2,2]	–	–
Assessment	Somewhat correct anatomy; significant black regions; exaggerated brightness	Mode collapse: undifferentiated grey brain regions	Mode collapse: inability to differentiate brain tissue from rest of scan	Mode collapse: noise, with learning of brain boundary	More correct intensity range and sharpness, though incorrect anatomy learning	Mode collapse: lack of anatomical differentiation, though improved intensity	More stable training; insufficient anatomy learning	Mode collapse: noise with minimal tumor region learning

Note: LR = Learning Rate; LR schedule was linear unless otherwise specified; LR warmup was kept at 500 training steps throughout; Noise loss was MSE unless otherwise specified; seg = modified segmentation mask. Epochs less than 100 were not specified, given early model failure. Letters a-g correspond to the output images of Figure 19

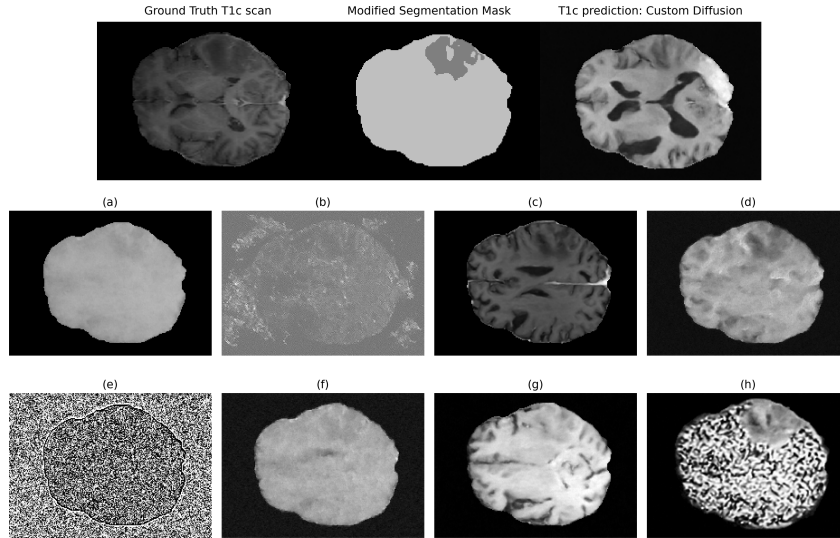


Figure 19: Examples of model outputs at various stages of training. The top right of the figure is the prediction of the best performing generator. The image on the top left is the ground truth T1c. The top-center image is the conditioned modified segmentation mask. Images labeled (a) through (h) represent predicted images following various input configurations and hyperparameter settings. Detailed training details are available in Appendix A3: Custom Diffusion, as well as descriptions of images a-h (Table 13).

6.14 A4: *StyleGAN2-ADA*

ARCHITECTURE This study employed the same architecture as was used by Tariq et al. (2023). The StyleGAN2-ADA generator is composed of a mapping network with 8 fully connected layers, as well as a synthesis network with 26 modulated convolutional layers (Karras et al., 2020). The mapping layers are responsible for translating the raw latent space Z into a new latent space of W , which correspond to general image features. Importantly, the learning rate of the mapping network is set to 100x lower than the synthesis network ($\text{lr_mul}=0.01$) to assure training stability, considering the significantly lower dimensionality of latent image representations.

The synthesis network consists of alternating upsampling and ToRGB layers, which are modulated by a vector from W (mapped from a sampled Z vector) rather than fixed weights, allowing the network to learn through individual images rather than a “one-size-fits-all” weight matrix. Weights from W are also normalized to prevent exploding gradients. The ToRGB connections function to generate a “partial image,” allowing information to be more stably passed on the fully connected layers. In addition, gaussian noise is added to feature maps at each layer, thereby allocating the generation of global structures to W and the generation of fine-grained detail to injected noise. A representation of the mapping and synthesis networks are shown in Figure 20.



Figure 20: A representation of the StyleGAN2-ADA generator architecture.

The discriminator consists of 8 resolution blocks, consisting of 24 convolutional layers preceded by a single convolutional layer responsible for translating from 3-channel RGB image space to the discriminator feature space. It also consists of a minibatch standard deviation layer, which measures batch diversity, helping the discriminator detect mode collapse (Figure 21).

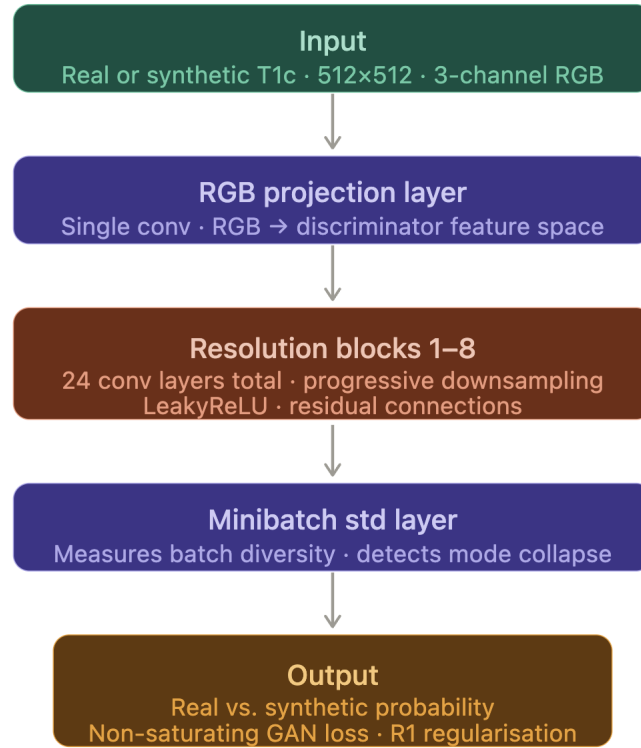


Figure 21: A representation of the StyleGAN2-ADA discriminator architecture.

TRAINING While Tariq et al. (2023) implemented the model with TensorFlow, this study elected to use the PyTorch version due to dependency issues. Pretrained weights from the AFHQ wild 512 × 512 dataset were used (Choi et al., 2020), as Tariq et al. (2023) used these weights to train a model with the second best FID/KID values at a permissible resolution of 512 × 512.

The model was trained using the PyTorch implementation, and was run for a total of 2,260 kimg. The generator at 1,880 kimg was ultimately chosen, as it had the lowest calculated FID value. No hyperparameter adjustments were conducted.

APPENDIX B

6.15 Error Analysis

Subsequent analysis was conducted at the tumor label level (NCR, Edema, ET) rather than the merged subregion level (TC, WT, ET). The cross-validated BraTS training set is referred to as CV in text, and "training" in figures and tables.

FP and FN rates FN, FP, and complete-miss (Dice = 0) rates were higher in the BraTS test set compared to the other datasets (Figure 11). This was observed across all tumor regions, but was most pronounced in small tumors. For small NCR regions, the BraTS test had an FN rate of 0.514 with a complete miss rate of 18.8%, much higher than for CV (0.250 and 6.7%) and for UCSF (0.149 and 3.5%). Although there were no complete misses for medium NCR, the FN and FP rate for BraTS test (0.234, 0.223) were substantially elevated compared to CV (FN: 0.085, FP: 0.056) and UCSF (FN: 0.053, FP: 0.041). Large NCR had a 1.1% complete miss rate for the BraTS test set, and a much higher FN rate (0.205) than for CV (0.054) and UCSF (0.086). FP rates were slightly more elevated for the BraTS test (0.080) than for UCSF and BraTS CV, both with FP < 0.05 (see Table 14).

Table 14: False negative (FN), false positive (FP), and complete miss rate by tumor subregion, size, and dataset. Complete miss is the percentage of cases where the predicted mask has zero overlap with the ground truth. n denotes the number of patients. Test refers to the BraTS test set, while train refers to the BraTS training set.

Subregion	Size	Dataset	n	FN	FP	Complete miss (%)
NCR	Small	Train	314	0.250	0.110	6.7
		Test	32	0.514	0.351	18.8
		UCSF	172	0.149	0.096	3.5
	Medium	Train	313	0.085	0.056	0.0
		Test	18	0.234	0.223	0.0
		UCSF	134	0.053	0.041	0.0
	Large	Train	323	0.054	0.042	0.0
		Test	189	0.205	0.080	1.1
		UCSF	80	0.086	0.039	1.2

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Table 14 (continued)

Subregion	Size	Dataset	n	FN	FP	Complete miss (%)
<i>Edema</i>	Small	Train	330	0.135	0.098	0.3
		Test	7	0.315	0.332	0.0
		UCSF	264	0.142	0.102	0.0
	Medium	Train	329	0.062	0.056	0.0
		Test	19	0.327	0.281	0.0
		UCSF	182	0.073	0.054	0.0
	Large	Train	340	0.043	0.039	0.0
		Test	225	0.148	0.124	0.0
		UCSF	53	0.069	0.046	0.0
<i>ET</i>	Small	Train	321	0.125	0.082	1.9
		Test	19	0.384	0.220	5.3
		UCSF	220	0.126	0.098	1.8
	Medium	Train	320	0.055	0.051	0.0
		Test	27	0.228	0.176	3.7
		UCSF	165	0.078	0.055	0.0
	Large	Train	330	0.043	0.042	0.0
		Test	196	0.103	0.081	0.0
		UCSF	37	0.079	0.061	0.0

For Edema, while there were no complete misses (save for 0.3% for CV), FN rates in the BraTS test dataset were much higher across all tumor sizes compared to BraTS CV and UCSF: small = 0.315 (CV: 0.135, UCSF: 0.142); medium = 0.327 (CV: 0.062, UCSF: 0.073); large = 0.148 (CV: 0.043, UCSF: 0.069). FP rates between datasets and tumor sizes followed a similar pattern.

Finally, small ET regions in the BraTS test set had a complete miss rate of 5.3% and FN = 0.384, compared to 1.9% and FN = 0.125 in CV, and 1.8% and FN = 0.126. Medium ET again showed elevated FN in the test set (0.228) compared to CV (0.055) and UCSF (0.078), with a miss rate of 3.7% compared to no misses across CV and UCSF. Large ET showed no complete misses across datasets, but still had BraTS with an elevated FN rate (0.103) compared to CV (0.043) and UCSF (0.079).

Across all subregions and tumor sizes, the FP and FN rates of UCSF remained consistently close to those of CV, while scores for BraTS test were notably more elevated, especially in small NCR.

Tumor size The BraTS test set contained considerably fewer small and medium tumor cases per region than either the CV or UCSF datasets. For instance, the NCR region of the test set contained 13.4% small, 7.5% large, and 79.1% large tumors – whereas each region was roughly equally represented in the CV dataset. UCSF had more small tumors than CV – for example in edema, with 52.9% small, 36.5% medium, and 10.6% large, again compared to roughly equal representation in CV (Figure 12).

Image Quality Statistics The distributions of both SNR and CNR were highly similar between the CV, test, and UCSF datasets (Figures 22 and 23).

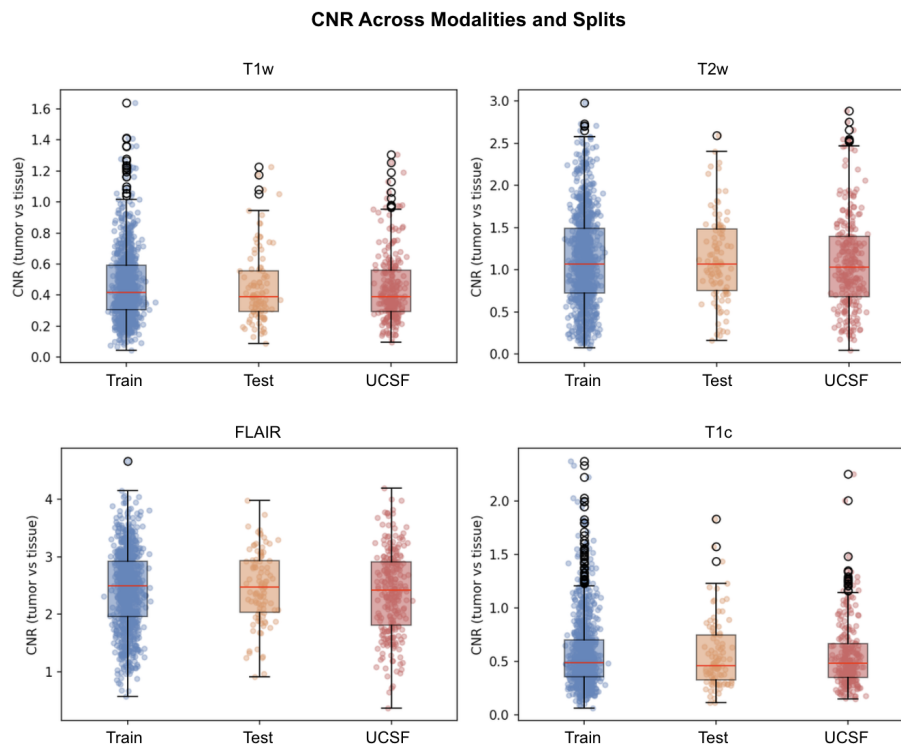


Figure 22: CNR values per scan between datasets.

However, cases of segmentation failure (i.e., individual patients with Dice < 0.3) were associated with significantly increased CNR and SNR in T2w and FLAIR scans within datasets.

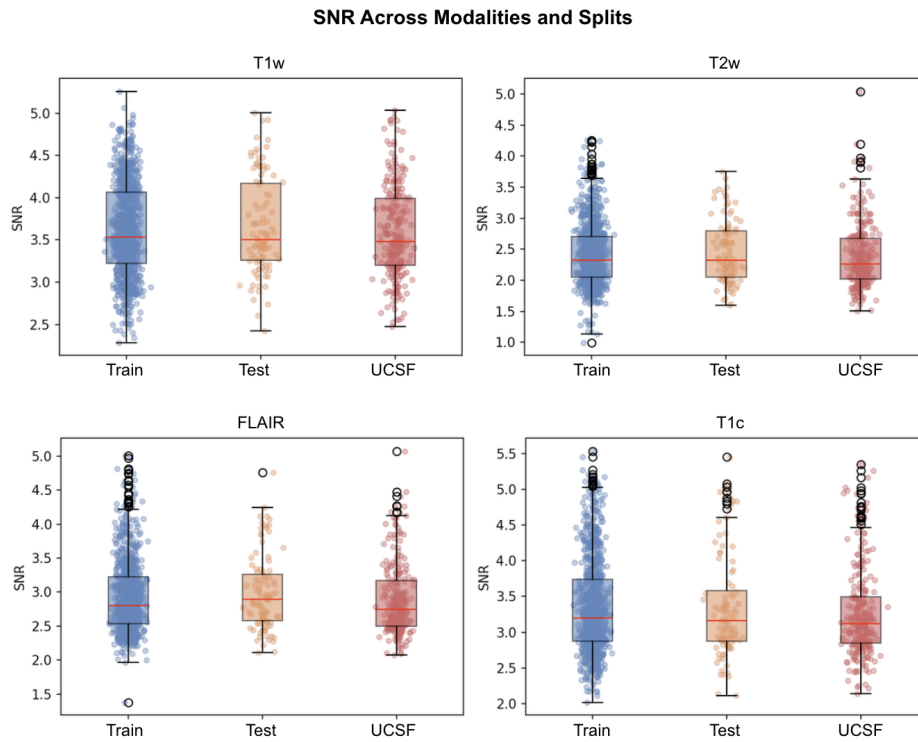


Figure 23: SNR values per scan between datasets.

Table 15 represents an exhaustive list of the SNR and CNR comparisons for individual scans between failed and passing segmentation cases. Effect sizes are reported (W_1), as well as significance values (Mann-Whitney p -values).

Table 15: Image quality metrics (SNR and CNR) for segmentation failures (Dice < 0.3) vs. passing cases (Dice ≥ 0.3) across all datasets and tumor subregions. Values are mean $\pm$ standard deviation. W_1 denotes the Wasserstein distance between the failed and passing distributions. p -values are from two-tailed Mann-Whitney U tests; significant results ($p < 0.05$) are shown in bold. There were no failed cases for mean Dice scores in the BraTS training set.

Subregion	Modality	Metric	Failed (mean $\pm$ sd)	Passing (mean $\pm$ sd)	W_1	p
BraTS Training Set (974 patients, train + val)						
	T1W	SNR	3.588 $\pm$ 0.503	3.629 $\pm$ 0.552	0.090	0.7438
	T2W	SNR	2.456 $\pm$ 0.451	2.415 $\pm$ 0.499	0.082	0.5310
	FLAIR	SNR	3.006 $\pm$ 0.578	2.919 $\pm$ 0.515	0.136	0.1760

NCR

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Table 15 (continued)

Subregion	Modality	Type	Failed (mean $\pm$ sd)	Passing (mean $\pm$ sd)	W_1	p
	T1C	SNR	3.611 ± 0.789	3.330 ± 0.645	0.281	0.0456
	T1W	CNR	0.542 ± 0.304	0.459 ± 0.222	0.111	0.1315
	T2W	CNR	0.950 ± 0.608	1.145 ± 0.551	0.225	0.0123
	FLAIR	CNR	2.357 ± 0.587	2.447 ± 0.677	0.140	0.3423
	T1C	CNR	0.588 ± 0.410	0.573 ± 0.321	0.084	0.6948
<i>Edema</i>	T1W	SNR	3.457 ± 0.465	3.623 ± 0.551	0.296	0.6731
	T2W	SNR	2.450 ± 0.341	2.417 ± 0.502	0.239	0.8271
	FLAIR	SNR	2.679 ± 0.323	2.928 ± 0.520	0.294	0.5334
	T1C	SNR	3.264 ± 0.192	3.347 ± 0.663	0.376	0.8235
	T1W	CNR	0.336 ± 0.075	0.464 ± 0.227	0.150	0.4374
	T2W	CNR	1.013 ± 0.789	1.128 ± 0.555	0.430	0.8126
	FLAIR	CNR	1.996 ± 0.542	2.437 ± 0.675	0.488	0.3960
	T1C	CNR	1.064 ± 0.617	0.575 ± 0.329	0.499	0.3344
<i>ET</i>	T1W	SNR	3.665 ± 0.754	3.624 ± 0.547	0.224	0.8003
	T2W	SNR	2.710 ± 0.445	2.406 ± 0.497	0.312	0.0047
	FLAIR	SNR	3.432 ± 0.811	2.914 ± 0.508	0.518	0.0088
	T1C	SNR	3.590 ± 0.941	3.342 ± 0.655	0.360	0.3757
	T1W	CNR	0.530 ± 0.214	0.463 ± 0.227	0.080	0.1687
	T2W	CNR	1.515 ± 0.533	1.118 ± 0.553	0.400	0.0035
	FLAIR	CNR	2.157 ± 0.750	2.447 ± 0.670	0.322	0.0583
	T1C	CNR	0.612 ± 0.276	0.575 ± 0.328	0.077	0.4141
BraTS Test Set (251 patients)						
<i>mean</i>	T1W	SNR	4.060 ± 0.542	3.586 ± 0.541	0.480	0.1771
	T2W	SNR	2.905 ± 0.321	2.385 ± 0.515	0.565	0.0432
	FLAIR	SNR	3.713 ± 0.497	2.876 ± 0.515	0.842	0.0133
	T1C	SNR	3.247 ± 0.088	3.266 ± 0.634	0.409	0.5706
	T1W	CNR	0.615 ± 0.125	0.444 ± 0.218	0.215	0.0899
	T2W	CNR	1.015 ± 0.609	1.089 ± 0.561	0.293	0.6912
	FLAIR	CNR	1.810 ± 0.572	2.388 ± 0.718	0.596	0.1961

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Table 15 (continued)

Subregion	Modality	Type	Failed (mean $\pm$ sd)	Passing (mean $\pm$ sd)	W_1	p
	T1C	CNR	0.408 $\pm$ 0.154	0.569 $\pm$ 0.315	0.179	0.3684
NCR	T1W	SNR	3.764 $\pm$ 0.593	3.561 $\pm$ 0.528	0.206	0.0916
	T2W	SNR	2.562 $\pm$ 0.530	2.369 $\pm$ 0.514	0.201	0.0328
	FLAIR	SNR	3.221 $\pm$ 0.649	2.837 $\pm$ 0.486	0.384	0.0004
	T1C	SNR	3.409 $\pm$ 0.650	3.251 $\pm$ 0.632	0.161	0.1145
	T1W	CNR	0.508 $\pm$ 0.270	0.437 $\pm$ 0.206	0.075	0.2296
	T2W	CNR	1.078 $\pm$ 0.618	1.102 $\pm$ 0.550	0.129	0.4561
	FLAIR	CNR	2.173 $\pm$ 0.889	2.419 $\pm$ 0.686	0.341	0.0611
	T1C	CNR	0.582 $\pm$ 0.346	0.563 $\pm$ 0.309	0.048	0.8527
Edema	T1W	SNR	3.308 $\pm$ 0.336	3.597 $\pm$ 0.545	0.317	0.2195
	T2W	SNR	2.510 $\pm$ 0.417	2.388 $\pm$ 0.517	0.207	0.4723
	FLAIR	SNR	2.982 $\pm$ 0.353	2.884 $\pm$ 0.525	0.233	0.3932
	T1C	SNR	3.053 $\pm$ 0.187	3.270 $\pm$ 0.635	0.339	0.6264
	T1W	CNR	0.717 $\pm$ 0.198	0.441 $\pm$ 0.214	0.285	0.0078
	T2W	CNR	0.850 $\pm$ 0.551	1.093 $\pm$ 0.561	0.289	0.2544
	FLAIR	CNR	1.572 $\pm$ 0.695	2.397 $\pm$ 0.711	0.825	0.0259
	T1C	CNR	0.837 $\pm$ 0.747	0.562 $\pm$ 0.296	0.343	0.9735
ET	T1W	SNR	3.674 $\pm$ 0.693	3.582 $\pm$ 0.532	0.206	0.9726
	T2W	SNR	2.961 $\pm$ 0.602	2.356 $\pm$ 0.493	0.625	0.0010
	FLAIR	SNR	3.427 $\pm$ 0.476	2.854 $\pm$ 0.503	0.589	0.0008
	T1C	SNR	3.565 $\pm$ 0.672	3.251 $\pm$ 0.628	0.331	0.1107
	T1W	CNR	0.539 $\pm$ 0.222	0.439 $\pm$ 0.216	0.111	0.1291
	T2W	CNR	1.303 $\pm$ 0.568	1.079 $\pm$ 0.558	0.239	0.1689
	FLAIR	CNR	2.037 $\pm$ 0.609	2.401 $\pm$ 0.720	0.377	0.0903
	T1C	CNR	0.571 $\pm$ 0.326	0.565 $\pm$ 0.312	0.094	0.9554
UCSF Dataset (501 patients)						
	T1W	SNR	3.484 $\pm$ 0.564	3.438 $\pm$ 0.562	0.134	0.9125
	T2W	SNR	2.114 $\pm$ 0.272	2.056 $\pm$ 0.275	0.104	0.6090
	FLAIR	SNR	2.558 $\pm$ 0.115	2.514 $\pm$ 0.190	0.093	0.4247

mean

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Table 15 (continued)

Subregion	Modality	Type	Failed (mean $\pm$ sd)	Passing (mean $\pm$ sd)	W_1	p
	T1C	SNR	3.118 ± 0.104	2.925 ± 0.311	0.237	0.0896
	T1W	CNR	0.438 ± 0.428	0.445 ± 0.265	0.174	0.3536
	T2W	CNR	0.935 ± 0.687	0.852 ± 0.506	0.266	0.9341
	FLAIR	CNR	1.574 ± 0.787	2.050 ± 0.583	0.481	0.2627
	T1C	CNR	1.216 ± 0.669	0.538 ± 0.298	0.677	0.0029
NCR	T1W	SNR	3.490 ± 0.636	3.494 ± 0.572	0.120	0.8625
	T2W	SNR	2.137 ± 0.199	2.032 ± 0.272	0.114	0.0419
	FLAIR	SNR	2.532 ± 0.150	2.499 ± 0.193	0.056	0.2013
	T1C	SNR	2.960 ± 0.285	2.894 ± 0.322	0.083	0.2519
	T1W	CNR	0.551 ± 0.353	0.407 ± 0.225	0.145	0.0672
	T2W	CNR	1.257 ± 0.629	0.800 ± 0.473	0.457	0.0002
	FLAIR	CNR	2.174 ± 0.562	2.027 ± 0.598	0.158	0.2497
	T1C	CNR	0.672 ± 0.341	0.495 ± 0.245	0.177	0.0018
Edema	T1W	SNR	3.268 ± 0.490	3.441 ± 0.562	0.230	0.4657
	T2W	SNR	2.004 ± 0.229	2.057 ± 0.276	0.125	0.8533
	FLAIR	SNR	2.451 ± 0.134	2.515 ± 0.189	0.102	0.5358
	T1C	SNR	2.995 ± 0.176	2.927 ± 0.311	0.149	0.6878
	T1W	CNR	0.250 ± 0.123	0.447 ± 0.267	0.197	0.1561
	T2W	CNR	0.474 ± 0.322	0.856 ± 0.509	0.391	0.1573
	FLAIR	CNR	1.517 ± 0.413	2.052 ± 0.582	0.547	0.1299
	T1C	CNR	0.816 ± 0.629	0.540 ± 0.296	0.332	0.7162
ET	T1W	SNR	3.158 ± 0.267	3.485 ± 0.570	0.349	0.0300
	T2W	SNR	2.183 ± 0.233	2.037 ± 0.273	0.148	0.0117
	FLAIR	SNR	2.526 ± 0.105	2.502 ± 0.190	0.073	0.2670
	T1C	SNR	3.055 ± 0.244	2.906 ± 0.319	0.162	0.0093
	T1W	CNR	0.590 ± 0.298	0.413 ± 0.229	0.183	0.0017
	T2W	CNR	1.278 ± 0.505	0.803 ± 0.482	0.479	<0.0001
	FLAIR	CNR	2.263 ± 0.605	2.025 ± 0.601	0.316	0.0238
	T1C	CNR	0.782 ± 0.482	0.497 ± 0.247	0.291	0.0011